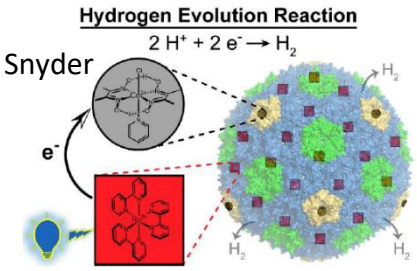
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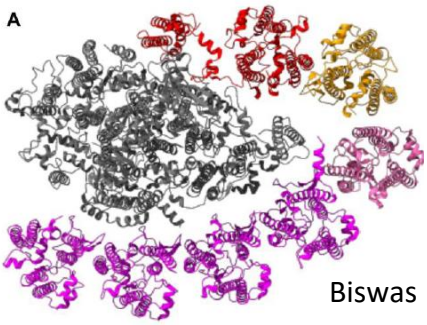
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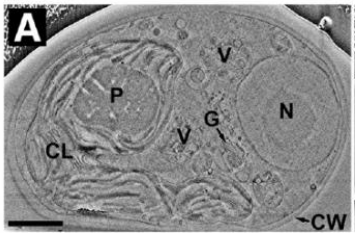
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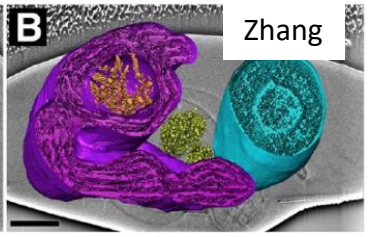


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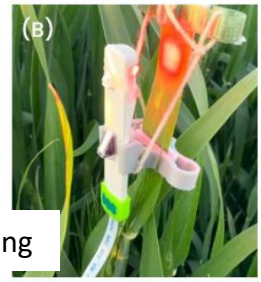
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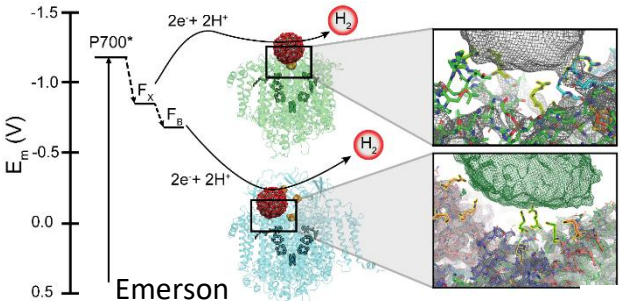


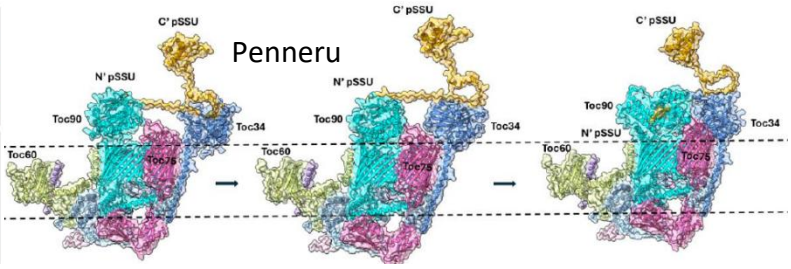
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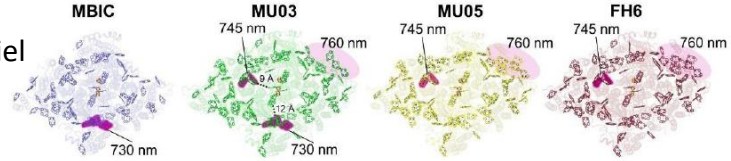


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Program

Friday, October 23

4:00 - 6:00 pm	**Registration & Poster Hanging**
6:00 pm	**Dinner**
7:30 pm	**Introduction and Welcome**
7:40 - 8:20 pm	**Keynote Lecture**: Laurie Leonelli (University of Illinois Urbana-Champaign): *Overlooked in the Understory: Managing Light Responses in the Shade*
8:20 - 8:40 pm	Erika Serravalle (University of Florida): *Substitutions in the pD1 C-terminal extension modulate proteolytic processing and photosynthetic stress tolerance*
8:40 - 9:00 pm	Sree Kavya Penneru (University of Tennessee Knoxville): *Dynamic GTPase Dimerization Coordinates the TOC Translocon*
8:00 - 9:20 pm	Swarnali Mukherjee (Oklahoma State University): *Understanding CO_2 uptake and proton pumping in cyanobacterial Photosynthetic Complex 1*
9:30 pm	**Mixer and Poster Session**

Saturday, October 24

7:00 am	**Breakfast**
9:00 - 9:40 am	**Keynote Lecture**: Kevin Redding (Arizona State University): *Insights into long-range electron transfer from the heliobacterial reaction center*
9:40 - 10:00 am	Christopher Gisriel (University of Wisconsin-Madison): *Multiple Routes to Red-Shifted Chlorophyll* d*-Based Photosynthesis*
10:00 - 10:20 am	Sabiha Tamanna (Georgia State University): *Probing the Reaction Center and Long-Range Electrochromism in Far-Red Photosystem I: A*

Comparative Study of Chroococcidiopsis thermalis *and* Fischerella thermalis

10:20 - 10:40 am **Coffee Break**

10:40 - 11:00 am Hasan Tuncer (Purdue University): *Electronic Structure and Hydrogen-Bond Control of O-O Bond Formation in Photosystem II*

11:00 - 11:20 am Christian Brininger (University of Wisconsin-Madison): *Quantifying Site-Specific Radiation Sensitivities within Photosystem II*

11:20 - 11:40 am Audrey Short (Argonne National Laboratory): *Investigating Fundamental Interactions Involved in Photosystem I-Driven Electron Transport*

11:40 - 12:00 pm Luisa Maria Sandoval Calle (Purdue University): *Constructing accurate Force Fields for (Bacterio)chlorophylls via ab initio calculations*

12:00 - 12:20 pm Christian Durbin (Michigan State University): *State Transitions in the Photoprotective Orange Carotenoid Proteins Probed by Molecular Dynamics Simulations*

12:30 pm **Lunch**

1:30 pm **Group Photo and Hike**

Afternoon Free

4:00 - 6:00 pm **Poster Session**

6:00 pm **Dinner**

7:30 - 7:50 pm Ru Zhang (Donald Danforth Plant Science Center): *Investigating Heat Induced Chloroplast Structural Changes by Multiscale Cryo-Volume Electron Microscopy and Cryo-Electron Tomography in the Green Alga* Chlamydomonas reinhardtii

7:50 - 8:10 pm Mahipal Rao (University of Tennessee Knoxville): *Light-driven Spatiotemporal Thylakoid Membrane Remodeling in Cyanobacterium* Chroococcidiopsis *TS-821*

8:10 - 8:30 pm	Jon Nguyen (Michigan State University): *Simulating Photosystem I-Ferredoxin Binding and Downstream Acceptor Specificity in* Chlamydomonas reinhardtii
8:30 - 8:50 pm	Mitchell Ticoras (Michigan State University): *Probing Photosynthetic Resiliency in Plants With a Metabolically Engineered Shikimate Pathway*
8:50 - 9:10 pm	Petra Fromme (Arizona State University): *Cryo-EM Structure Determination of PSI-LHCI and a PSII-antenna complex from the CAM plant* Opuntia Ellisiana *(Prickly pear cactus)*
9:15 pm	**Poster Session**

Sunday, October 25

7:00 am	**Breakfast**
9:00 - 9:20 am	Berkley Walker (Michigan State University): *What does photorespiration have to do with photosynthetic energy balancing?*
9:20 - 9:40 am	Maria Santos Merino (Michigan State University): *Phosphorylated sugars as signaling molecules in cyanobacteria*
9:40 - 10:00 am	Maxwell Harman (Michigan State University): *Genetic Dissection of Photosynthetic Oscillations Reveals ATP Synthase's Central Coordination of Carbon/Light Reaction Balance*
10:00 - 10:20 am	Harvey Hou (Alabama State University): *Tri-Institutional (NLR-WashU-ASU) Cyanobacterial CUREs at Alabama State University*
10:20 - 10:40 am	Panel: *AI and its responsible use in the photosynthesis community*
10:40 - 11:00 am	**Coffee Break**
11:00 am	**Awards Presentations**

Abstracts

Oral abstracts are listed first, in chronological order. Poster abstracts are listed alphabetically by the last name of the presenting author.

Overlooked in the Understory: Managing Light Responses in the Shade

Laurie B. Leonelli[1,2,3], Jake Harris[1,2], Komal Attri[2], Giulia Lauria[2], Atinder Singh[1], Kithmee de Silva[3], Lucia Arce Cubas[2], Elena Pelech[2], and Matthew D. Brooks[2,3]

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A plant growing in the field must contend with both light extremes: at the top of the canopy, excess absorbed light energy can damage the photosynthetic machinery, while at the bottom of the canopy, insufficient light can limit productivity. Plants manage these challenges through various molecular strategies that allow them to adjust their responses to the light environment, from rapidly cycling xanthophylls to regulating the expression of photosynthesis-related genes. Today's talk examines photosynthetic responses to shade through two complementary studies in soybean (*Glycine max*) and sorghum (*Sorghum bicolor*). In soybean, we discuss a newly identified enzyme, LEP, involved in the lutein epoxide cycle, a xanthophyll cycle known to regulate light use in low-light conditions. In sorghum, we investigate photosynthetic inefficiencies observed in the deep shade of sorghum canopies, with the goal of identifying the regulatory components responsible for this seemingly maladaptive response. Together, these studies provide complementary perspectives on the molecular and physiological mechanisms underlying plant responses to low light. Integrating these approaches can improve our understanding of how different crop species respond to shaded environments and may help identify targets for engineering efforts to improve photosynthetic performance under canopy-specific light conditions.

Substitutions in the pD1 C-terminal extension modulate proteolytic processing and photosynthetic stress tolerance

Erika A. Serravalle[1], Haohao Zhao[1], Zhenxiong Jiang[1], and Xin Wang[1]

[1]*Department of Microbiology and Cell Science, Institute of Food and Agricultural Sciences, University of Florida, Gainesville, FL 32611, USA.*

Oxygenic photosynthesis depends on the continual repair of Photosystem II (PSII). This repair cycle is driven largely by turnover of the reaction center D1 protein, which is highly susceptible to photodamage. In *Synechococcus elongatus* PCC 7942, D1 is synthesized as a precursor (pD1) bearing a 16-residue C-terminal extension that is sequentially cleaved by the lumenal protease CtpA to generate mature D1 and enable assembly of the oxygen-evolving Mn_4CaO_5 cluster. Despite the centrality of this step to PSII biogenesis and repair, how individual residues within this transient extension affect proteolytic processing and photosynthetic stress tolerance remains unclear. Here, we investigated three substitutions within the pD1 C-terminal extension, L353F, I358N, and H359N, previously identified in our directed evolution screen and associated with enhanced growth under salt stress. To determine how sequence variation affects the CtpA-pD1 interaction, we quantified their binding affinities using microscale thermophoresis and found that each substitution altered CtpA binding. We then characterized the thermodynamic basis of CtpA-pD1 recognition using isothermal titration calorimetry, revealing that binding was entropically driven. To determine the functional relevance of these binding differences, we next quantified CtpA-mediated cleavage of wild-type and variant pD1 substrates *in vitro*. Bulky, hydrophobic substitutions at residue 353 enhanced proteolytic processing of the pD1 C-terminal peptide, whereas I358N and H359N slowed processing. Despite these opposing effects on processing kinetics, each variant conferred a growth advantage over wild-type pD1 under combined salt and light stress when expressed at neutral site III. This suggests additional mechanisms, such as assembly efficiency or interactions with assembly factors, contribute to the observed stress-tolerant phenotypes in the variants. Together, these findings establish the pD1 C-terminal extension as a functional determinant of photosynthetic stress tolerance and, given the conservation of D1 maturation across oxygenic phototrophs, identify it as a promising target for engineering stress-resilient photosynthesis.

Dynamic GTPase Dimerization Coordinates the TOC Translocon

Sree Kavya Penneru[1], Rajan Lamichhane[1] and Barry D. Bruce[1,2]
[1]*Biochemistry, Cellular and Molecular Biology, University of Tennessee, Knoxville, TN-37916.*
[2]*Genome Science and Technology, University of Tennessee, Knoxville, TN-37916.*

Photosynthetic capacity depends on the assembly of a predominantly nuclear-encoded chloroplast proteome, which requires precise coordination from precursor recognition through translocation. At the chloroplast surface, the TOC receptor GTPases Toc34 and Toc159/Toc90 recognize transit peptides and initiate import, but the molecular motions linking recognition to translocation remain unclear. Here, we use single-molecule Förster resonance energy transfer (smFRET), Martini coarse grained simulations, and chemical cross-linking to define the in vitro oligomerization and conformational dynamics of their purified G-domains. The Toc34 G-domain adopts a predominantly dimeric basal state that is destabilized by transit peptides and nonhydrolyzable GTP analogs. In contrast, transit-peptide binding promotes Toc34–Toc159/Toc90 G-domain heterodimerization, which is further remodeled by nucleotide state. An engineered XTP-specific Toc34 enables independent control of its nucleotide occupancy while preserving the GTP specificity of Toc159/Toc90. smFRET reveals a dynamic heterodimeric ensemble whose state populations and transition kinetics depend on the nucleotide occupancy of each receptor. Selective XTP loading of Toc34 accesses conformations not detected under conventional GTP conditions, demonstrating distinct contributions from the two G-domains. Together, these findings transform the TOC receptors from static docking factors into a dynamic, nucleotide-regulated molecular switch and suggest how transit-peptide recognition triggers coordinated rearrangements that move precursors toward productive engagement with the chloroplast translocation machinery.

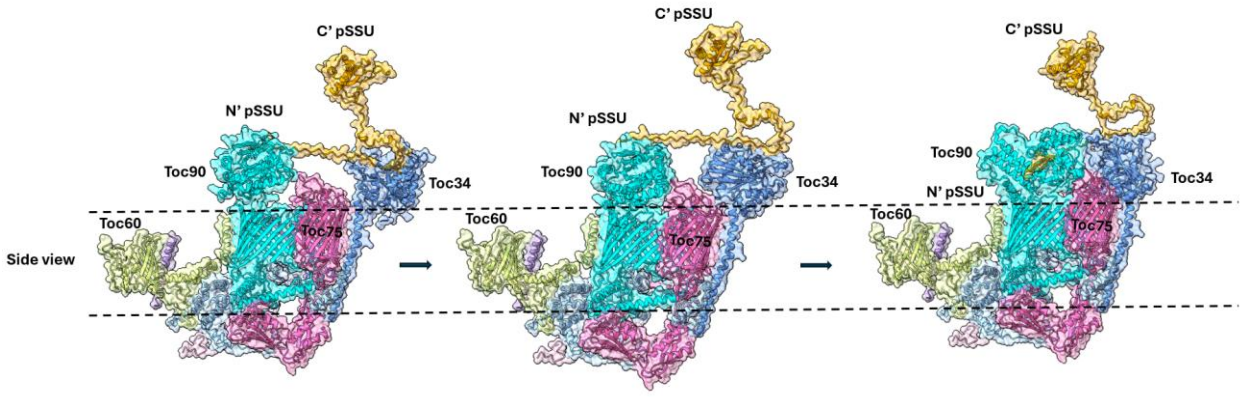


Figure 1. Proposed model of Toc34–Toc90 heterodimer formation during precursor protein import. A hybrid structural model of the TOC complex was constructed by combining the cryo-EM structure of its membrane-embedded core (PDB 7XZJ) with AlphaFold 3–predicted cytosolic domains of Toc34 and Toc90 and a predicted full-length precursor of the small subunit of Rubisco (pSSU). The three side views illustrate a proposed sequence of receptor rearrangements: initial contact between the pSSU transit peptide and the TOC complex (left), an intermediate state (middle), and formation of the Toc34–Toc90 G-domain heterodimer (right). Toc60, Toc90, Toc75, Toc34, and pSSU are shown in lime green, cyan, pink, blue, and orange, respectively. The black dotted line marks the membrane boundary. Each structure is shown as a transparent molecular surface overlaid on a cartoon representation.

Understanding CO_2 uptake and proton pumping in cyanobacterial Photosynthetic Complex 1.

Swarnali Mukherjee,[1] Robert L Burnap[1]
[1]*Oklahoma State University, Stillwater, USA-* swarnali.mukherjee@okstate.edu

To survive under low CO_2 conditions, cyanobacteria have evolved a carbon-concentrating mechanism (CCM) that accumulates intracellular inorganic carbon (Ci) and enhances CO_2 fixation by Rubisco. Central to this system are specialized NAD(P)H dehydrogenase-like complexes (NDH-1; photosynthetic Complex I) that couple electron-coupled proton transfer and CO_2 uptake. The NDH-1_4 isoform contains the extrinsic CupB subunit and catalyzes CO_2 hydration at an unusual Zn-containing active site formed at the CupB–NdhF4 interface.
Unlike canonical carbonic anhydrases, which catalyze reversible CO_2 /HCO_3^- interconversion, the CupB active site is proposed to drive unidirectional, energized CO_2 hydration, allowing cells to maintain a large cytoplasmic bicarbonate pool. Walker et al. (Biochim Biophys Acta Bioenerg. 2024 1865:149149) proposed that residues from NdhF4, particularly Arg37 and Glu114, facilitate unidirectionality by transiently stabilizing the bicarbonate product and promoting its release by extracting and pumping a proton away from the Arg-Bicarbonate-Glu adduct. To test this model, we developed a molecular genetic complementation system and generated targeted substitutions at these residues.
Arg37 substitutions impaired NDH-1-dependent CO_2 uptake and growth without disrupting PSII activity. Growth defects were rescued by increased bicarbonate availability in the growth medium, while SbtA upregulation and preliminary MIMS data suggest diminished bicarbonate accumulation and CO_2 retention in the mutants. Together, these results are consistent with the product trapping hypothesis that Arg37 contributes to the directionality of CO_2 hydration and intracellular bicarbonate retention. Disruption of this residue may permit increased reverse CO_2 /HCO_3^- interconversion and compromise the cytoplasmic C_i pool.

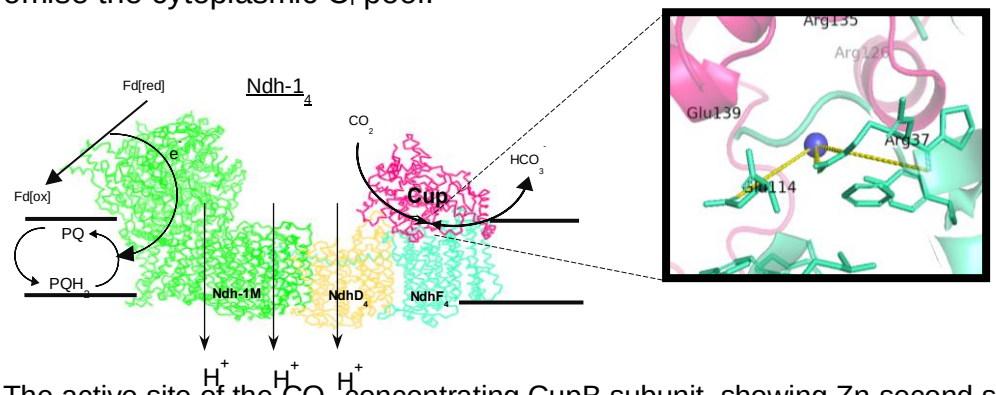


Figure1. The active site of the CO_2 concentrating CupB subunit, showing Zn second sphere NdhF4-Arg37 and NdhF4-Glu114 residues.

This work is supported by the United States Department of Energy, Office of Basic Energy Sciences, DE-FG02-08ER15968

Insights into long-range electron transfer from the heliobacterial reaction center

Kevin REDDING[1], Jesse GRANSTROM[1], Kristen KNAPPENBERGER[1], Kathryn TOMAH[1], Adam PIRNIA[1], Shreya SHAW[1], Abhishek SINGHAROY[1], Dmitry MATYUSHOV[1]

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The photochemical reaction centers (RCs) found on this planet appear to have evolved only once and subsequently diversified into the extent versions found now. The heliobacterial reaction center (HbRC) may be the best model for a structurally simple (being essentially a homodimer of a single polypeptide) and enzymatically flexible photo-enzyme. Although the HbRC is a Type I RC that typically reduces ferredoxins (or other soluble low-potential electron carriers), our work indicates that in some conditions the HbRC can use the endogenous menaquinone as an alternate electron acceptor, and that this may be controlled by the binding of an isoprenyl-phosphate in the presumed quinone-binding site. A combined computational and experimental project has identified features of the protein that are important for facilitating electron transfer either to the quinone or to the F_X [4Fe-4S] cluster on the cytosolic side of the protein that reduces ferredoxins. The long-range (18 Å) electron transfer to the iron-sulfur cluster is predicted by semi-classical calculations to be in the microsecond timescale, and yet we find a sub-nanosecond decay time, indicating that there are important contributors to this transfer step that are currently unknown to us. A hypothesis will be presented to explain this perplexing observation. We have used various inhibitors that block electron transfer at different steps to distinguish in vivo between reduction by the HbRC of quinone vs ferredoxin to determine the relative importance of each cyclic electron transport pathway to the cell under different conditions. Key results on the components of the electron transport pathways will be summarized.

Multiple Routes to Red-Shifted Chlorophyll *d*-Based Photosynthesis

Christopher J. Gisriel[1], Himanshu S. Mehra[1], Nikki Cecil M. Magdaong[1], David A. Flesher[2], Gaozhong Shen[1], Nikea J. Ulrich[3], Christian M. Brininger[1], Scott R. Miller[3], Dariusz M. Niedzwiedzki[4]

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The cyanobacterium *Acaryochloris marina* is unusual in that it uses chlorophyll *d*, which is red-shifted relative to the more common chlorophyll *a*. Different *A. marina* strains exhibit striking differences in their absorption and fluorescence spectra, arising primarily from variation in Photosystem I (PSI). Here, we compare PSI from several *A. marina* strains to understand how these spectral differences arise and how they influence excitation energy transfer. Spectroscopic measurements reveal distinct combinations of low-energy chlorophyll states among strains, producing a continuum of spectral phenotypes (Fig. 1). Cryo-EM structures of PSI at ~1.8 Å resolution identify differences in the local environments of chlorophylls that may generate these states, indicating that the low energy sites does not arise from a single conserved structural motif. Phylogenetic analyses further suggest that this spectral diversity evolved through modular variation and reassortment of PSI components. Despite these substantial differences in pigment energetics, transient absorption measurements show remarkably similar overall excitation energy transfer and trapping dynamics among the strains. These findings reveal considerable flexibility in how chlorophyll *d*-containing PSI can tune its energetic landscape while maintaining efficient energy transfer, providing insight into photosynthesis near the low-energy limit of oxygenic photosynthesis.

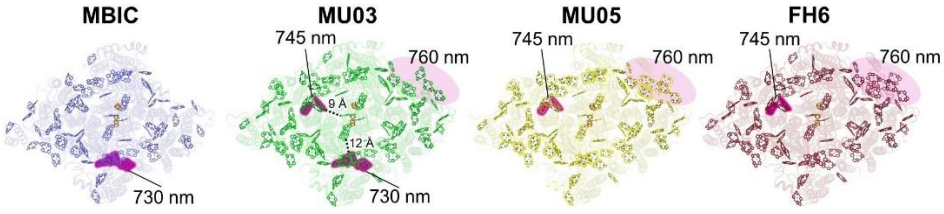


Figure 1. Locations of candidate low-energy sites in PSI from each *A. marina* strain. The distances of candidate sites associated with fast fluorescence features are shown in the MU03 structure only.

Probing the Reaction Center and Long-Range Electrochromism in Far-Red Photosystem I: A Comparative Study of *Chroococcidiopsis thermalis* and *Fischerella thermalis*

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The incorporation of chlorophyll f (Chl f) into Photosystem I (PSI) extends cyanobacterial photosynthesis beyond the canonical 700 nm red limit. While peripheral antenna Chl f sites are well established, the presence and functional role of Chl f within the core electron-transfer chain remain debated. Here, we compare the low-temperature (77 K) light-induced absorption difference spectra ($P_{700}^{+} - P_{700}$) of trimeric PSI from *Chroococcidiopsis thermalis* PCC 7203 (*C. thermalis*) and *Fischerella thermalis* PCC 7521 (*F. thermalis*) grown under white and far-red light (FRL). In both species, FRL acclimation produces a distinct negative bleaching band near 748–752 nm that is absent in white-light complexes. This shared feature supports the presence of a redox-active Chl f cofactor at the $A_{-1\mathrm{B}}$ site directly participating in primary charge separation. Above 720 nm, the positive difference baseline in both organisms reflects the broad absorption of the photo-oxidized primary donor (P_{700}^{+}). Furthermore, both species display a broad electrochromic band shift near 800 nm, attributable to the long-range electrostatic influence of P_{700}^{+} acting on the coupled $B37/B38$ Chl f terminal dimer. These results indicate that core photochemical and low-energy trapping mechanisms are conserved across diverse far-red-acclimated cyanobacteria.

Electronic Structure and Hydrogen-Bond Control of O-O Bond Formation in Photosystem II

Hasan Tuncer[1], Pavel Pokhilko[1, 2] and Yulia Pushkar[1*]

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Photosystem II catalyzes light-driven water oxidation at its Mn_4CaO_5 oxygen-evolving complex, yet the electronic rearrangements governing O-O bond formation remain incompletely understood. A recently proposed mechanism identifies O3 and O6 as coupling partners and suggests that hydrogen bonding between His337 and O3 regulates substrate reactivity.[1] Complementary work demonstrates that competing self-consistent-field solutions, electronic instabilities, and method-dependent redistribution of electrons between manganese and oxygen complicate conventional broken-symmetry descriptions of this cluster.[2] Together, these findings motivate a systematic investigation of how the protein environment influences the electronic configurations involved in oxygen formation.

We propose to examine the relationship between His337 hydrogen bonding, metal-oxygen electronic character, and O-O coupling using broken-symmetry density functional theory. Beginning with B3LYP, targeted initial guesses will be generated using localized open-shell orbitals to explore competing electronic solutions for S_3 reactant models and candidate O3-O6 and O5-O6 peroxide intermediates. Orbital-stability analysis will assess converged solutions, while localized-orbital analysis will characterize the distribution of unpaired electrons across manganese and oxygen centers. For selected configurations, homotopy continuation and natural difference orbitals will reveal how changes in the electronic-structure approximation affect metal-oxygen electron redistribution. Comparisons across matched His337 hydrogen-bonding configurations will test whether the predicted preference for O3-O6 coupling persists across these electronic descriptions. This study aims to clarify the electronic role of His337 in substrate activation and assess the robustness of mechanistic predictions for biological water oxidation.

References

[1] Pushkar, Y. New Water Oxidation Mechanism in Photosystem II Resolves Major Experimental Controversies. *Angew. Chem. Int. Ed.* **2026**, *65*, e1930324. DOI: 10.1002/anie.1930324.

[2] Pokhilko, P.; Pushkar, Y. Algorithm for Locating SCF Solutions Reveals Conceptual Limitations of the Broken-Symmetry Approach Applied to the Oxygen-Evolving Complex. *ChemRxiv* **2026**, version 1, preprint. DOI: 10.26434/chemrxiv.15006047/v1.

Quantifying Site-Specific Radiation Sensitivities within Photosystem II

Christian Brininger[1], Anoushka Kuckreja[1], Christopher J. Gisriel[1]

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Single particle cryo-electron microscopy (cryo-EM) has proven to be a powerful tool for investigating proteins and their cofactors at high resolution. However, recent studies show that certain features within photosystem II (PSII), such as the oxygen-evolving complex (Mn_4CaO_5), are altered during traditional cryo-EM data acquisition and analysis pipelines (Kato et al. (2021), Gisriel et al. (2022)). This introduces discrepancies between structural information and spectroscopic data, raising questions about what other features are meaningfully altered during imaging. Here, we implement a low-dose data acquisition scheme and analysis pipeline to reconstruct PSII structures across fractional electron exposures (1 to 40 $e^-/ Å^2$) at high resolution (2.25 to 1.93 Å). Using this framework, we systematically track the dose dependent impacts of electron radiation damage on the oxygen evolving complex, disulfide bonds, bound pigments, and overall protein secondary/ tertiary structure. Our findings provide site specific insights into radiation sensitivity and will help to establish best-practice workflows for resolving the native states of metallocofactors and other protein features.

Investigating Fundamental Interactions Involved in Photosystem I-Driven Electron Transport

Audrey H. Short[1], Lisa M. Utschig[1]

[1]*Chemical Sciences and Engineering Division, Argonne National Laboratory, Lemont, IL;* *ashort@anl.gov*

Photosynthesis harnesses energy from the sun, driving electron transport and carbon fixation. In the photosynthetic Z-scheme, the final electron acceptor is $NADP^+$. PSI powers the formation of NADPH by transferring a light-generated electron to ferredoxin (Fd), which subsequently interacts with Fd-$NADP^+$ reductase (FNR) to catalyze the reduction. Here, we investigated how species-dependent organization of PSI and its electron acceptors influences electron transfer to both native partners and an abiotic catalyst. By comparing $NADP^+$ reduction in thylakoids from three different eukaryotic species (*Spinacia oleracea, Chlorella vulgaris, Scenedesmus obliquus*), we observed significant enhancement of NADPH production with excess Fd/FNR only in spinach thylakoids. To test whether these differences arise from accessibility of the PSI acceptor side, we used platinum nanoparticles (PtNPs) as bioinorganic probes of charge transfer between PSI and its native electron acceptors. Negatively charged mercaptosuccinic acid-capped PtNPs self-assemble at positively charged regions of PSI near the Fd binding site, as shown in cryo-EM structural studies, and catalyze H_2 production. Among the thylakoid biohybrids, only spinach samples produced H_2, suggesting that PSI in algal thylakoids may be embedded in tightly associated supercomplexes that restrict access to both native electron transfer partners and PtNPs. To distinguish effects of native membrane organization from intrinsic PSI reactivity, we then isolated PSI from all three species and examined charge transfer directly to PtNPs. These isolated PSI biohybrids also showed species-dependent behavior, with spinach and *S. obliquus* showing substantial H_2 production. Together, these results show that species-dependent organization strongly influences charge transfer from PSI, affecting interactions with both native redox partners and abiotic catalysts.

Constructing accurate Force Fields for (Bacterio)chlorophylls via ab initio calculations

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In recent years, there has been significant research into biofuels generated through photosynthesis as a renewable energy source. However, current production methods are inefficient and fail to meet current demand. Therefore, it is crucial to study photosystems both experimentally and computationally to understand and optimize their functionality. Traditionally, the structure and dynamics of biological systems have been investigated using molecular modeling, which relies on classical force fields (FFs). However, the reliability of predictive modeling depends heavily on the ability of FFs to accurately represent the potential energy function. Chlorophylls (Chls), the primary pigments in plants, play a vital role as a solar energy-absorbing pigments and as electron donors in the electron transport chain, converting solar energy into chemical energy. This study aims to develop and validate accurate FFs for Chl and Bacteriochlorophylls (BChl) using the Q-Force software tool. The parameters for bonds, angles, rigid dihedrals and inversion dihedrals were derived from the quantum-mechanical (QM) Hessian, the atomic charges are fitted using the QM electrostatic potentials, while Lennard-Jones parameters were obtained from the parent force field. Flexible dihedral parameters were derived from QM potential energy surface scans to reproduce the multiple-minima behavior of the dihedral potentials. The resulting force fields are validated by comparing QM and molecular mechanics potential energy surfaces, as well as against previously reported FFs from the literature. Molecular dynamics simulations of the Fenna-Matthews-Olson (FMO) complex and water-soluble chlorophyll protein (WSCP) have also been performed to evaluate the FFs in photosynthetic systems. These new force fields are expected to improve the quality and reliability of computational simulations of photosynthetic systems and further our understanding of photosynthesis and its potential applications.

State Transitions in the Photoprotective Orange Carotenoid Proteins Probed by Molecular Dynamics Simulations

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Photosynthetic organisms must balance light harvesting against photodamage when exposed to varying light conditions. Nonphotochemical quenching (NPQ) is a major photoprotective mechanism to dissipate excess absorbed light energy as thermal energy. In cyanobacteria, orange carotenoid protein (OCP) mediates this process, quenching excess light captured by the phycobilisome. In response to blue-green light, OCP converts from its inactive state, OCP^O, to its active state, OCP^R, a process characterized by translocation of the bound ligand, canthaxanthin (CAN), and by dissociation of the N-terminal (NTD) and C-terminal domains (CTD). Current hypotheses of OCP activation pose that structural elements of OCP and activation step sequentiality regulate OCP activation, but it remains unknown how these affect the energetics of OCP activation. By individually biasing the change in ligand binding pose or the change in protein conformation, we aim to identify the mechanistic regulation of OCP activation using molecular dynamics simulations. Pilot steered molecular dynamics simulations indicate that CAN translocation is much less energetically costly compared to NTD-CTD dissociation and may lower the energy barrier for the separation of the two OCP domains. This is supportive of the experimentally determined fast timescale (ns-μs) in which CAN translocation occurs compared to domain dissociation (ms). Further, we aim to measure how the N-terminal extension and CTD residues Y201 and W288 regulate the equilibrium of OCP states by calculating free energy profiles along simulated reaction coordinates for CAN translocation and domain dissociation and across OCP mutants. This study will deliver mechanistic insight into the regulation of orange carotenoid protein activation.

Investigating Heat Induced Chloroplast Structural Changes by Multiscale Cryo-Volume Electron Microscopy and Cryo-Electron Tomography in the Green Alga *Chlamydomonas reinhardtii*

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Photosynthesis is critical for plant biomass production, but it is frequently challenged and compromised by various stresses in nature, including high temperatures. Heat stress affects various cellular processes, and photosynthesis is one of the most heat sensitive functions. Investigating heat effects on chloroplast structures at three-dimension (3D) and nanometer scales is essential to understand photosynthesis under high temperatures, but it remains largely underexplored. Cryo-Volume Electron Microscopy (cryo-vEM) and Cryo-Electron Tomography (cryo-ET) are emerging molecular imaging techniques for visualization of the 3D spatial organization of cellular structures embedded in vitreous ice. Cryogenic imaging methods preserve cells in their near-native environment without traditional chemical fixation, heavy metal staining, or resin embedding. Cryo-vEM allows visualization of nearly all cellular structures within an entire cell volume in 3D at ~5 nm resolution; while cryo-ET focuses on specific thin lamellae with areas of interest to reveal detailed cellular structures at 0.5-1 nm resolution. Both cryo-imaging method complement each other, and they can largely advance our understanding of stress-induced cellular structure changes (especially chloroplasts) and help make novel discoveries that cannot be achieved using traditional methods with 2D views and/or low resolutions. Our recent study showed that the eukaryotic, model green alga *Chlamydomonas reinhardtii* (*C. reinhardtii*) had differentially reduced photosynthetic rates during 24-hour moderate (35°C) or acute (40°C) high temperatures. Based on our transmission electron microscopy analysis, *C. reinhardtii* cells exposed to 40°C had disorganized thylakoids. We hypothesize that high temperatures with different intensity and duration may affect chloroplast structures, particularly thylakoid structures and pyrenoids for carbon concentrating mechanisms, which alter photosynthetic efficiencies under high temperatures. We are investigating ultrastructural changes of chloroplasts in heat-treated *C. reinhardtii* cells under native status *in situ* by employing 3D cryo-vEM and cryo-ET. The information revealed with unprecedented precision and resolution can largely improve our understanding of photosynthesis under stressful conditions and help improve stress tolerance in photosynthetic cells. Our high-resolution results and developed pipeline of cryo-vEM and cryo-ET will also be valuable for researchers who study other photosynthetic cellular processes under different stressful conditions.

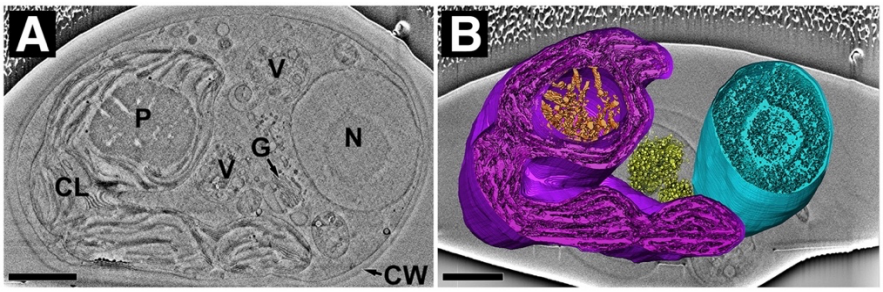


Figure. 1. 3D cryo-vEM results from Chlamydomonas, cited from (Czymmek et al. MPMI, 2023).

Light-driven Spatiotemporal Thylakoid Membrane Remodeling in Cyanobacterium *Chroococcidiopsis* TS-821

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Conversion of solar energy into chemical energy is the primary driver for most life on Earth. In photosynthetic organisms, this process occurs in dynamic, multilamellar thylakoid membranes (TMs) that harbor the photosynthetic protein complexes. Cyanobacteria, which are the evolutionary precursors of chloroplasts, exhibit highly variable TM architectures, ranging from no discernible membranes (e.g., in *Gloeobacter violaceus*) to wavy or irregular shapes and, in some cases, concentric or partial multilamellar structures. Interestingly, many well-studied cyanobacteria (e.g., *Synechocystis* and *Thermosynechococcus*) display a more uniform, highly ordered TM arrangement. Here, we investigate a thermotolerant, non-heterocyst-forming *Chroococcidiopsis* TS-821, which possesses a reticulated, fascicular TM system. *Chroococcidiopsis* TS-821 can transition from a dimeric photosystem I (PSI) to a tetrameric PSI under high light. Structural analysis of this tetrameric PSI reveals a concave lumen surface, which may drive pronounced changes in thylakoid curvature under high light. Although much is known about the photosynthetic processes of cyanobacteria, the dynamic spatial organization of TMs—especially in living cells—remains poorly understood. To address this gap, we employed non-destructive, high-resolution small-angle neutron scattering (SANS) on actively growing TS-821 cells to examine how TM organization responds to different growth light levels. Our SANS data reveal differences in TM organization corresponding to changes in light intensity, potentially stemming from PSI oligomerization, as previously shown by our lab. To further elucidate these findings, we plan to correlate the SANS data with electron microscopy, thereby clarifying the structural underpinnings of TM reorganization under varying light conditions. This work enhances our understanding of TM organization in cyanobacterium known to express PSI in tetrameric form. It also provides insights into how different cyanobacterial classes may share or diverge in TM architectures, shedding light on the evolutionary trajectories that shape photosynthetic membrane systems.

Simulating Photosystem I-Ferredoxin Binding and Downstream Acceptor Specificity in *Chlamydomonas reinhardtii*

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Ferredoxin (Fdx) is the terminal electron acceptor from Photosystem I (PSI) and partitions these electrons across various cellular processes, including carbon fixation, nitrogen and sulfur assimilation, and chlorophyll biosynthesis. *Chlamydomonas reinhardtii* encodes as many as 12 chloroplast Fdx isoforms, which appear to be specialized for particular pathways. Fdx1/Fdx2 are involved in NADPH production, nitrite and sulfite assimilation, while Fdx3/Fdx6 are involved in chlorophyll biosynthesis. How this specificity is encoded remains unknown. Most ferredoxins compete at the PSI acceptor site for electrons. Thus far, we have simulated 9 Fdx isoforms bound to PSI. Utilizing AI tools such as AlphaFold3, the 9 Fdx isoforms all sit at the stromal hump of PSI, where PsaC,PsaD, and PsaE form the binding pocket, similar to what is seen in a recent *C. reinhardtii* cryo-EM structure (PDB: 7WZN). PSI remains stable throughout simulations. Fdxs 1-5 and 9 maintain stability, while Fdxs 6-8 displayed substantial deviations due to their extended loop domains. Cofactor geometry is preserved throughout, with Fe-S cluster edge-to-edge distances between many Fdxs and PSI between 8-12 Å. These Fe-S cluster distances are well within the range of efficient electron transfer. Fdx7 is the sole outlier, with Fe-S cluster distances ranging between 18-28 Å. Contact number shows that 7 Fdx isoforms remain bound throughout their simulations, with Fdx7 and Fdx9 showing a dramatic dropoff in contact number. The 9 Fdx isoforms show varying degrees of stability when bound to PSI, though contain close to ideal Fe-S distances for electron transfer. Extending this investigation to nitrite/sulfite assimilation and chlorophyll biosynthesis pathways, we scored AlphaFold3 models with PROtein binDIng enerGY prediction (PRODIGY) and the Adaptive Poisson-Boltzmann Solver (APBS), Fdx1 is predicted to have stronger affinity with sulfite reductases than Fdx2, with Fdx1 and Fdx2 being indistinguishable with nitrite reductases. Fdx3 and Fdx6 models with aerobic cyclases show a glaring distinction in affinity, with Fdx6 having >~2.5X the affinity towards aerobic cyclases for chlorophyll biosynthesis. Poisson-Boltzmann electrostatics reveal electrostatic steering, as every Fdx binding face is negative, and nitrite/sulfite reductase and aerobic cyclase faces are strongly positive. These results map individual Fdxs onto their acceptors and identify binding stability, cofactor distances for efficient electron transfer, and isoform-specific extensions as determinants of specificity.

Probing Photosynthetic Resiliency in Plants With a Metabolically Engineered Shikimate Pathway

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With the discovery of newer, quicker, and more reliable molecular biology technologies, our ability to probe how plants balance photosynthetic output with metabolic demand is improving. Recently, *Arabidopsis thaliana* mutants called *suppressor of tyra2*, or *sota*, mutants have been isolated that lack normal feedback regulation of the well-studied shikimate pathway. As a result, these *sota* mutants hyperaccumulate aromatic amino acids, the downstream products of the shikimate pathway, without showing any obvious growth phenotypes. We use these promising mutations to ask whether the rewiring of sink metabolism affects photosynthetic output and resiliency in both *Arabidopsis thaliana* and *Glycine max*.

First, in *sota* arabidopsis strains, we probed steady state carbon dioxide assimilation rates and assessed whether these mutants exhibit changes in metabolic electron demand. Moreover, we asked if *sota* plants can maintain robust photosynthesis when challenged with fluctuating environmental conditions like changes in light intensity and temperature. Current gas exchange and fluorometry results indicate that these *sota* arabidopsis mutants remain largely unchanged in terms of photosynthetic capacity or resiliency despite their aromatic amino acid accumulation phenotype.

Interestingly, when soybean *sota* mutants were also tested for changes in aromatic amino acid content, carbon assimilation, and growth in greenhouse conditions a different set of phenotypes emerged. While these mutants do hyperaccumulate aromatic amino acids, carbon assimilation is not maintained near wild-type levels. Additionally, *sota* lines showed aberrations in growth and development timeline unlike what was seen in the arabidopsis mutants. As such, these data act as evidence that engineering of the shikimate pathway and the rewiring of metabolism downstream of photosynthesis has species-specific effects on both photosynthesis itself and plant development. However, the results in arabidopsis do support that such engineering efforts may be achieved without stark costs to plants, hinting that future engineering projects aimed at industrial deployment could be feasible.

Cryo-EM Structure Determination of PSI-LHCI and a PSII-antenna complex from the CAM plant *Opuntia Ellisiana* (Prickly pear cactus)

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Photosynthetic organisms have evolved diverse modifications to their photosynthetic machinery in response to fluctuating light intensities and dynamic environmental conditions. The structural diversity of Photosystem I (PSI) and Photosystem II (PSII) reflects how variations in antenna composition and pigment organization have evolved to optimize light capture and protect against photodamage. Crassulacean Acid Metabolism (CAM) is an adaptive photosynthetic pathway that evolved in higher plants to thrive under arid and high-light environments. CAM plants minimize water loss by fixing CO_2 at night into malic acid, which is stored in vacuoles and later decarboxylated during the day to support photosynthesis. Given that these plants endure extreme light intensities yet maintain photosynthetic efficiencies comparable to C_3 and C_4 species, structural insights may uncover adaptive modifications and specialized energy-transfer pathways that help the cactus plants adapt to these extreme conditions.

We have successfully determined a high-resolution structure of the PSI-LHCI complex from a CAM plant solved at 2.2 Å with a red shifted fluorescence spectrum, marking one of the first structural characterizations of this complex in any cactus species. In my talk I will show this first structure of the Photosystem I-Light Harvesting Complex I (PSI-LHCI) from a cactus, the CAM plant *Opuntia ellisiana,* determined by cryogenic electron microscopy (cryo-EM) at 2.2 Å and also show first preliminary results of a cryo-EM structure from PSII-antenna complexes from the cactus. Exciting features of the PSI-LHCI structure are that the complex is red shifted and also that it features a much tighter coupling of the LHCI "belt" with the core of Photosystem I. In my talk I will discuss how these functional and structural features of PSI-LHCI complex from a desert cactus may allow it to survive under the extreme desert conditions and deepen our understanding of PSI evolution to adapt the photosynthetic machinery to arid, high-light environments.

What does photorespiration have to do with photosynthetic energy balancing?

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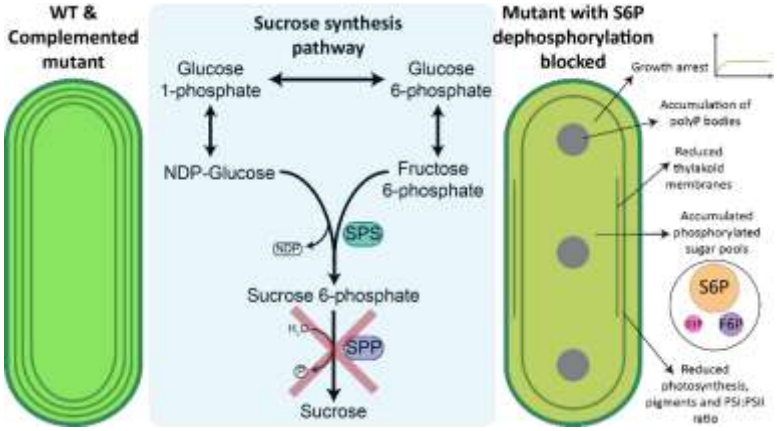
Photosynthesis operates through the coordinated balance of energy capture, carbon assimilation, and metabolic demand. This balance is struck between the capture of energy (in terms of absolute abundance and ATP:NADPH ratios) and the consumption of the energy through downstream metabolism. When this balance is perturbed, either ROS damage can occur due to unquenched energy, or the system operates in a less-efficient state. In many photosynthetic organisms, photorespiration consumes 30-40% of ATP and reducing equivalents produced from the light reactions. Given this large energy demand, what is the role of photorespiration in energy balancing? Our work determines that photorespiration's role needs to be considered in the network context by drawing on physiological measurements, metabolic flux analyses, systems biology, and evolutionary approaches. For example, meta-analyses of isotopically non-stationary metabolic flux studies revealed that the impact of photorespiration on ATP:NADPH demand is buffered by increased starch synthesis when photorespiration is minimized. Further, theoretical and measured flux analyses demonstrated that photorespiratory intermediates frequently exit the canonical pathway to support amino acid, one-carbon, and other biosynthetic metabolism, substantially altering the energetic consequences of Rubisco oxygenation. These alternative routes can shift whole-cell ATP:NADPH demand, establishing photorespiration as a dynamic contributor to energy balancing rather than a fixed energetic cost. Our biophysical and physiological experiments further challenged the prevailing view that photorespiration primarily acts as a photoprotective electron sink under non-stress conditions. Manipulating photorespiratory flux in tobacco demonstrated that reductions in photorespiration led to feedback limitations associated with ATP synthase regulation and decreased photosystem II efficiency. These findings indicate that the large ATP demand of photorespiration is crucial for maintaining ATP synthase conductivity and preventing energetic bottlenecks in the chloroplast. Under heat stress, however, photorespiration contributes to maintaining electron flow and protecting the photosynthetic apparatus when carbon assimilation becomes restricted, highlighting distinct roles in both routine energy balancing and environmental stress responses. To understand how photosynthetic systems adapt to sustained changes in photorespiratory demand, we combined transcriptomics, proteomics, and metabolic modeling to examine acclimation to altered oxygen environments. These studies revealed rapid and coordinated reallocation of metabolic capacity to restore photosynthetic performance when photorespiratory flux changes, supporting the concept that photorespiration is deeply integrated with wider carbon and nitrogen metabolism. Ongoing genetic studies have identified mutants impaired in transitions between photorespiratory states, providing evidence that nitrogen remobilization and amino group supply represent critical links connecting photorespiration, metabolism, and energy balance. Collectively, our findings support a revised framework in which photorespiration is a flexible component of photosynthetic energy regulation that ties together metabolic and photosynthetic energy conversion and demand.

Phosphorylated sugars as signaling molecules in cyanobacteria

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Plants tightly regulate photosynthetic capacity and metabolism through sugar-based signaling. Key metabolite pools act as proxies for systemic energy status, sensed by transducer proteins that trigger downstream adjustments in resource allocation. In plants, the metabolites involved in sugar signaling are either upstream intermediates of the sucrose biosynthetic pathway (*i.e.*, glucose 6-phosphate and glucose 1-phosphate (G1P)), or components of analogous sugar biosynthetic pathways (*i.e.*, trehalose 6-phosphate). While this mechanism is well characterized in plants, the sensors and metabolites cyanobacteria use to balance photosynthesis with metabolism remain largely unknown. To investigate the potential role of sucrose metabolism substrates and sucrose 6-phosphate (S6P) as potential regulatory metabolites, we genetically engineered the model cyanobacterium *Synechococcus elongatus* PCC 7942 to block the sucrose synthesis pathway at the level of S6P dephosphorylation. After activating the first step of the sucrose synthesis pathway catalyzed by SPS (sucrose phosphate synthase), the engineered strain showed growth arrest, decreased photosynthetic activity, a lower PSI:PSII ratio, and reduced pigments. All these phenotypes were accompanied by an accumulation of large polyphosphate bodies and a reduction in the number of thylakoid layers. Targeted metabolomics was used to quantify intracellular pools of phosphorylated sugars. As expected, S6P levels increased 10-fold in this mutant, accompanied by elevated G1P and fructose 6-phosphate (F6P). Lastly, expression of enzymes that dephosphorylate S6P (sucrose phosphate phosphatases; SPPs) prevented all the phenotypes described above. Together, these results suggest phosphorylated sugars act as signaling molecules in cyanobacteria—coordinating photosynthesis and metabolism and/or crosstalking with stress pathways—with future work needed to disentangle the individual roles of the three identified phosphorylated sugars.

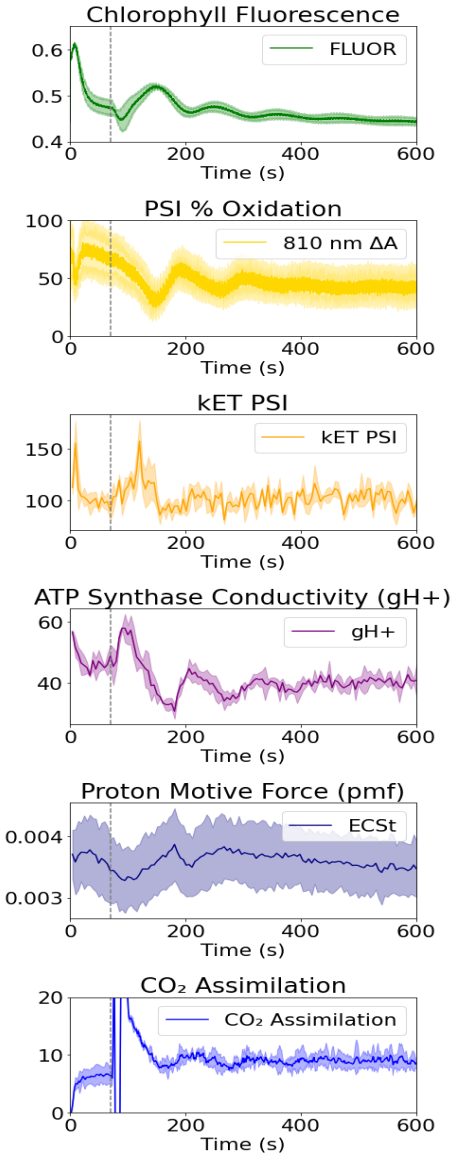


Phosphorylated sugars coordinates photosynthesis and metabolism in cyanobacteria.

Genetic Dissection of Photosynthetic Oscillations Reveals ATP Synthase's Central Coordination of Carbon/Light Reaction Balance

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Rapid increases in photosynthetic flux induce characteristic dampening oscillations spanning chlorophyll fluorescence to carbon fixation. Despite being a classical observation, the mechanisms coordinating these disparate processes have remained unresolved. Here, we pair established *Arabidopsis thaliana* mutants with high-resolution, multi-signal optical spectroscopy and gas exchange to monitor thylakoid proton motive force, ATP synthase activity, and electron transfer kinetics during carbon dioxide step-changes. We demonstrate that these classical transient oscillations are driven by dynamic inhibition of ATP synthase. Upon sudden exposure to elevated carbon dioxide, rapid acceleration of the Calvin-Benson cycle depletes chloroplastic inorganic phosphate as it becomes sequestered in CBC intermediates. This acute phosphate shortage depresses ATP synthase conductivity, causing a transient buildup of proton motive force that throttles electron transport through Cytochrome b_6f via photosynthetic control. Simultaneous measurement of Photosystem I reduction kinetics reveals that transient electron accumulation stems from these shifting upstream electron supply dynamics rather than a downstream acceptor-side limitation. Furthermore, genetic knockouts confirm that secondary energy balancing mechanisms, including non-photochemical quenching and cyclic electron flow, are not required to generate or dampen the oscillations. These results identify ATP synthase as a central coordinator balancing thylakoid energy production with carbon demand, while demonstrating how classical gas exchange can be paired with advanced spectroscopic approaches to dissect complex, real-time metabolic feedback loops in vivo.

Figure #1. Multi-signal photosynthetic oscillations in response to high CO_2. *A. thaliana* Col-0 leaves were adapted to 1500 µmol/m^2/s light at 420 ppm CO_2. After a 90 second baseline measurement period, CO_2 was increased to 2000 ppm (gray vertical dashed line). Optical measurements made with an integrated diode emitter array spectrophotometer/fluorometer (IDEA-spec) integrated into an LI-6800 for gas exchange and environmental control. PSI k_{ET}, $g_{H+,}$ and ECS_t were measured via the dark interval relation kinetics (DIRK) method every five seconds. N=4, shading = +/- standard deviation.

Tri-Institutional (NLR-WashU-ASU) Cyanobacterial CUREs at Alabama State University

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To enhance the capability and competitiveness of the underrepresented students to enter Ph.D. programs and the job market, the tri-institutions, including the National Laboratory of the Rockies (NLR), Washington University in St. Louis (WashU), and Alabama State University (ASU), have teamed up to create a comprehensive research and training program at Alabama State University. In this talk, I will share ideas and practices for using course-based undergraduate research experiences (CUREs) at Alabama State University through the tri-institutional cyanobacteria program. There are three levels in the training program: (1) teaching classroom and laboratory in a variety of classes, for example, general chemistry (CHE 141 and 142), forensic science survey (FRS 220), forensic chemistry (FRS 320), forensic toxicology (FRS 423), and senior capstone seminar (FRS 499). (2) Hands-on experience in working in the research laboratory at ASU by working on a weekly basis with graduates, postdoc researchers, and the PI. (3) Full summer independent research experience at NLR, WashU, and ASU for about 10 weeks to conduct fundamental research in photosynthesis and applications of cyanobacteria with world-leading scientists. The tri-institutional program includes CUREs modules: a professional ethics discussion, oral presentations on the literature critique, group experiments using research on cyanobacteria, written group reports in the format of a journal paper, and AI assignments on the research. In addition, we organized five annual research events, including All-Hands Meeting, CARE Workshop, Spring Symposium, Summer Symposium, and Fall Symposium. Students via CURE modules greatly benefit from a well-structured, year-round program with measurable outcomes, and build long-term, strong commitment and confidence in STEM professional development. This work was supported by the U.S. Department of Energy, SFEE program (DE-SC002567) and RENEW program (DE-SC0024702) and National Science Foundation (NSF MCB-2428256).

Posters

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Kelsey Dahlgren-Weinstock

Lauren Dome

Maximino Emerson

Michael Eyeson

Mengjie Fan

Frey Frenelle

Chris Garcia

Muhammad Haroon

Marcos Hendler

Jingcheng Huang

Kavan Jackson

Reneé Kimbro

Patryk Kuleta

Giulia Lauria

Chloe Leaman

Nikki Cecil Magdaong

Matt Martin

Linta Maruthurethu Biju

Andrea Meyer & Sivanujan Suthaharan

Keith Meyers

Saima Mirza

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Emily Nakayama

Michael Nelson

Julia Niedzwiedzka

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Lily Pumphrey

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Zachary Ryan

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Chetna Sharma

Swapnil Singh

Samuel Snyder

Viktoriia Stolyar

Praveena Suresh

Shimo Tang

Sham Vera

Shellie Wall

Chloe Wilke

Stenasia Wilson

Yuan Xu

Neetu Yadav

Madalyn Zagajowski

Cysteine Driven Charge Transfer Enables Photoprotection in Carotenoid Free Protein

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Fenna-Matthews-Olson (FMO) pigment protein is essential in transferring excitation energy from the chlorosome antenna to the reaction center in green sulfur bacteria. Although the FMO complex lacks carotenoids that typically guard photosynthetic systems against reactive oxygen species generated under excess light, it is unusually stable in aerobic environments.

This unsolved photoprotective behavior has led to various hypotheses. One idea is that the lowest-energy triplet state of the FMO's pigment lies below the energy of singlet oxygen, inhibiting the complex from sensitizing singlet oxygen in the first place. In 2016, Orf et al. [1] put forward a hypothesis that the FMO's photoprotection occurs from two cysteine residues located near bacteriochlorophyll (BChl) pigments 2 and 3. They proposed that the thiyl radicals formed by these cysteines under oxidative conditions act as acceptors for a rapid electron transfer (ET) from the excited BChl. This ET, followed by the back thiyl-to-BChl electron transfer, dissipates excitation on BChl pigments 2 and 3 nonradiatively.

On the other hand, apart from short-lived thiyl radicals, cysteines can also reversibly oxidize to sulfenic acid, which is found naturally throughout many protein environments. In this scenario, the fast ET could occur between the deprotonated sulfenic acid and BChl. In that case, the direction of electron flow would be inverted relative to the mechanism proposed by Orf et al.

Thus, we advanced to computationally prove the latter postulate. Our modeling approach combines classical molecular dynamics simulations followed by constrained quantum mechanics / molecular mechanics (QM/MM) geometry optimizations of photosynthetic pigments, and excited-state calculations with the polarizable embedding QM/EFP (EFP: Effective Fragment Potential) approach. We show that in the model systems in which the cysteine is computationally mutated to its deprotonated oxidized form (Cys-SO^{-}), Cys → BChl charge-transfer state is isoenergetic with the Q_y excitation for the lowest-energy BChl 3. On the other hand, the wild-type FMO and the models with protonated oxidized cysteine do not show low-energy CT states. The observed low-energy CT state appearing under oxidizing conditions offers an efficient route for dissipating excess excitation energy, thus limiting the generation of singlet oxygen by the FMO protein and protecting the bacterial reaction center from overloading.

[1] Orf, G. S.; Saer, R. G.; Niedzwiedzki, D. M.; Zhang, H.; McIntosh, C. L.; Schultz, J. W.; Mirica, L. M.; Blankenship, R. E. *PNAS* 2016, *113*, E4486–E4493. DOI: 10.1073/pnas.1603330113

Dynamic Photoprotection Under Stress: Distinct Acclimation Strategies in *Chlamydomonas priscui* and *C. reinhardtii*

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Photosynthetic organisms frequently experience fluctuating environmental conditions that challenge cellular energy balance and photosynthetic performance. Adjustments in photosynthetic activity to protect against damage and reestablish energy balance are known as photoacclimatory mechanisms. The implementation of the photoacclimatory mechanisms varies by organism, stress type, and stress duration, all of which alter the outcome of the response. Currently, crop yields are declining, in part due to a changing climate. Thus, identification of photoacclimatory strategies that promote optimal photosynthetic activity and growth under stress is an important strategy for improving crop stress resilience. Here, we compared short- and long-term acclimation to high light and salt stress in the Antarctic polyextremophile *Chlamydomonas priscui*, isolated from Lake Bonney in the McMurdo Dry Valleys, and the mesophilic model *Chlamydomonas reinhardtii*. Cultures were maintained under continuous growth conditions in photobioreactors (FMT150, Photon Systems Instruments), enabling assessment of photosynthetic performance and photoprotective responses during prolonged stress exposure (days). The two species exhibited distinct and stress-dependent acclimation responses. Under high light, both species initially showed a rapid reduction in PSII activity (Y[II]). However, after 48 hr, Y[II] recovered in the extremophile, *C. priscui*, while *C. reinhardtii* maintained a lower Y[II] throughout the treatment. Furthermore, high-light-treated *C. priscui* activated NPQ transiently, while *C. reinhardtii* exhibited minimal capacity to induce NPQ. Under high-salt stress, *C. priscui* similarly exhibited a rapid initial decline in Y[II] followed by a slow recovery. In contrast with high light exposure, salt-stressed cultures of *C. priscui* exhibited a slow increase in NPQ throughout the treatment. High salt-treated cultures of *C. reinhardtii* exhibited a slow decline in Y[II] over several days, followed by partial recovery. Together, these results show that acclimation across temporal scales is complex and is shaped by both the nature of the stressor and the organism's evolutionary history of exposure to that stress. Our findings highlight the dynamic regulation of photoprotective mechanisms during long-term acclimation and provide insight into how photosynthetic organisms maintain photosynthetic function under prolonged environmental stress.

3-Hydroxybutyrate as an Overflow Metabolite in *Synechocystis sp. PCC* 6803: Origin and Regulation

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Cyanobacteria can redirect excess fixed carbon into storage compounds and overflow metabolites in response to environmental stress (Sebesta et al 2024). One such metabolite, 3-hydroxybutyrate (3-HB), has been reported as an important energy regulatory mechanism in cyanobacteria and might has potential applications in sustainable bioproduction (Sebesta et al 2025). However, its metabolic origin in cyanobacteria remains unclear. Specifically, it is unknown whether extracellular 3-HB is generated through polyhydroxybutyrate (PHB) degradation or by direct removal of CoA from 3-hydroxybutyryl-CoA (3HB-CoA). This study aims to define the metabolic origin of 3-HB and identify the endogenous enzyme(s) responsible for its formation in Synechocystis sp. PCC 6803 under highlight stress. Preliminary results showed that wild-type cells produced detectable extracellular 3-HB, whereas the ΔphaAB mutant, which is unable to form 3HB-CoA, showed little to no detectable 3-HB. These findings indicate that 3HB-CoA formation is required for 3-HB production (Sebesta et al 2015). To distinguish between PHB-dependent and PHB-independent pathways, 3-HB production will be compared among wild-type, ΔphaAB, and ΔphaEC strains. Because ΔphaEC cannot polymerize 3HB-CoA into PHB, detection of 3-HB in this mutant would provide evidence for a direct 3HB-CoA → 3-HB pathway independent of PHB degradation. Extracellular 3-HB will be quantified by HPLC, while PHB and glycogen will be measured to evaluate changes in intracellular carbon partitioning. Candidate CoA-removing enzymes, including putative acyl-CoA thioesterases and hydrolases, will subsequently be investigated using CRISPR interference (CRISPRi). Identifying the enzyme responsible for 3HB-CoA hydrolysis would reveal a previously uncharacterized metabolic overflow mechanism in cyanobacteria and provide new targets for engineering photosynthetic microorganisms to enhance the production of 3-HB and other sustainable biochemicals. This work was supported by the U.S. Department of Energy, SFEE program (DE-SC002567) and RENEW program (DE-SC0024702), and in part by the National Science Foundation (NSF MCB-2428256).

Jacob Sebesta, Michael Cantrell, Eric Schaedig, Harvey J.M. Hou, Colleen Pastore, Katherine J. Chou, Wei Xiong, Michael T. Guarnieri1, and Jianping Yu* (2024) Polyphosphate kinase deletion increases laboratory productivity in cyanobacteria. Front. Plant Sci. 15:1342496.

Jacob Sebesta, Troy Paddock, Harvey J.M. Hou, Arnav Deshpande, Lieve Laurens, Katherine J. Chou, Michael T. Guarnieri, and Jianping Yu, "Wild-type Synechocystis sp. PCC 6803 secretes 3-hydroxybutyrate." 15th Workshop on Cyanobacteria, Nashville, Tennessee, June 5-8, 2025

Near-Atomic-Resolution Structure of Native Trimeric Photosystem I from the Thermophilic Cyanobacterium *Thermosynechococcus elongatus*

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Abstract: Photosystem I (PSI) is a large type I photosynthetic reaction center that catalyzes light-driven electron transfer in oxygenic photosynthesis. Here, we report a 2.18 Å cryo-electron microscopy (cryo-EM) structure of native trimeric PSI from the thermophilic cyanobacterium *Thermosynechococcus elongatus* (syn. *T. vestitus*), isolated from thylakoid membranes using styrene–maleic acid (SMA) copolymer without conventional detergent solubilization. The resulting PSI styrene–maleic acid lipid particle (PSI–SMALP) preserves the native trimeric reaction center at near-atomic resolution together with a substantial membrane-derived environment. Notably, portions of the SMA polymer are partially resolved as concentric belt-like densities surrounding the perimeter of the complex and positioned adjacent to the two leaflets of the retained thylakoid bilayer. This organization provides direct structural support for the widely proposed annular polymer-belt model of SMALPs and, to our knowledge, represents the first cryo-EM evidence directly localizing SMA polymer relative to the two leaflets of a membrane-protein-containing lipid particle. Independent biochemical analysis indicates retention of an extensive lipid population, estimated at approximately 800 lipid molecules per PSI trimer, with enrichment of the anionic sulfolipid, sulfoquinovosyldiacylglycerol (SQDG). Most of these associated lipids are dynamic and are therefore not individually resolved in the cryo-EM reconstruction. However, well-defined lipid densities are observed at selected protein surfaces and monomer–monomer interfaces, consistent with contributions to trimer stabilization and maintenance of local pigment and cofactor organization. Comparison with detergent-solubilized PSI reveals substantial differences in the membrane-associated environment while preserving the overall architecture of the reaction center and its extensive pigment–cofactor network. These structural observations complement ultrafast spectroscopic measurements showing altered, and in several components faster, excitation-energy dynamics in PSI–SMALPs relative to detergent-solubilized PSI, suggesting that retention of the surrounding membrane environment can influence PSI photophysical behavior. Together, these results demonstrate that detergent-free SMA solubilization is compatible with near-atomic-resolution structural analysis of native trimeric PSI while retaining a bilayer-like membrane environment. The structure provides a framework for examining the organization of PSI, its associated lipids and cofactors, and the surrounding membrane within a native-like particle, while direct visualization of the SMA belt provides structural support for the long-standing model of polymer organization in SMALPs.

Investigating Protein Environment Effects on Bacteriochlorophyll Excitation Energies in the Fenna–Matthews–Olson (FMO) Complex

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The Fenna–Matthews–Olson (FMO) pigment–protein complex is a well-studied light-harvesting system that mediates excitation energy transfer (EET) from peripheral antenna complexes to the reaction center in green sulfur bacteria. It is also known for its near-unity quantum efficiency of energy transfer, which can be attributed to pigment–pigment interactions shaped by evolutionary constraints within the local environment. Although molecular structures are available for FMO and other antenna complexes, understanding their electronic transitions and energy transfer dynamics has largely relied on spectroscopic measurements.

Energy transfer in FMO involves excitonic transitions among eight bacteriochlorophyll a (BChl a) pigments embedded in the protein environment of each of its three monomers. The resulting excitonic properties reflect the substantial influence of the local protein environment on the energetic landscape of the pigments. Our group recently developed a multiscale, first-principles, polarizable quantum mechanics/molecular mechanics (QM/MM) model that accurately reproduces the experimental absorption and circular dichroism spectra of FMO. A central component of this approach is the effective fragment potential (EFP) method, which demonstrated the importance of accounting for protein polarization when modeling these experimental features. In this work, we investigate how the protein environment influences BChl excitation energies and decompose the overall protein effect into contributions from specific amino acid residues responsible for substantial shifts in these energies. This decomposition identifies the structural and energetic mechanisms by which single-point mutations might affect excitonic interactions in FMO. Overall, our modeling provides structure-function relations for excitonic interactions and energy transfer in FMO.

Light-Induced Visible Difference Spectra of Photosystem I from Four Organisms

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Photosystem I (PSI) has a highly conserved reaction-center architecture, although the organization and spectral properties of its chlorophyll antenna vary among photosynthetic organisms. To examine how these differences are reflected in light-induced spectral changes, Fourier transform visible (FT-Vis) difference spectra were measured for purified PSI from *Thermosynechococcus elongatus*, *Synechocystis* sp. PCC 6803, *Synechococcus* sp. PCC 7002, and *Chlamydomonas reinhardtii*. Measurements were carried out under comparable conditions at room temperature and 77 K.

The spectra from all four PSI preparations are dominated by a bleaching feature near 700 nm associated with P700. Despite this common feature, differences are evident in the spectral line shapes, particularly in the chlorophyll Qy region and toward longer wavelengths. The position of the principal bleaching varies only slightly among the four PSI preparations, whereas considerably larger variation is observed in the long-wavelength zero crossing. At 77 K, additional features are resolved between approximately 660 and 700 nm, where electrochromic responses of chlorophylls surrounding the reaction center contribute to the difference spectrum. Differences are also observed in the near-infrared region associated with the P700 cation radical.

The measurements show that the characteristic P700-associated bleaching is largely maintained across the four PSI preparations, while other features of the light-induced difference spectrum vary among the organisms. These observations provide a common experimental framework for examining how the conserved PSI reaction center is spectroscopically influenced by its surrounding pigment environment.

Keywords: Photosystem I; P700; FT-Visible Spectroscopy; Light-Induced Difference Spectroscopy; Reaction Center; Chlorophyll; Low-Temperature Spectroscopy

Engineering phosphoglycolate phosphatase sequence to improve thermotolerance

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The conversion of sunlight into chemical energy is essential for plant survival, yet photosynthetic efficiency is limited by several biochemical constraints. One major limitation is the oxygenation reaction catalyzed by rubisco, which produces the toxic metabolite 2-phosphoglycolate. Photorespiration mitigates this toxicity by recycling 2-phosphoglycolate through the activity of the enzyme phosphoglycolate phosphatase (PGP). However, photorespiration is an energetically costly process that diverts carbon away from carbon fixation and carbohydrate production. Importantly, increasing temperature shifts rubisco specificity toward oxygen, thereby increasing photorespiratory flux. Previous work from our lab demonstrated that PGP activity increases in the heat-adapted species *Rhazya stricta*, suggesting that PGP may play an important role in maintaining photorespiratory flux under elevated temperatures and could represent a target for improving photosynthetic performance. Additional unpublished work indicates an increase in PGP catalytic efficiency that correlates with the temperatures a species is adapted to.

To investigate the enzyme features that impart improved catalytic efficiency with temperature, we are testing the kinetics of ancestral PGP inferred from either a thermophilic or mesophilic branch of extant species. Additionally, we generated site-directed mutants of *A. thaliana* PGP targeting amino acid sequences that exhibit high variability among plant species with differing levels of thermotolerance. Together, these approaches aim to identify sequence features that contribute to PGP function under elevated temperature and provide insight into the potential for engineering PGP activity to support increased photorespiratory flux.

Exploring the role of two IsiA homologs in *Anabaena* sp. PCC 7120

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Iron-stress-induced A (IsiA) protein serves as a chlorophyll *a*-binding antenna for photosystem I (PSI) under iron-deficient conditions in cyanobacteria. IsiA is best known for forming an 18-mer ring around a trimeric PSI. Recently, an unusual $IsiA_7$-PSI_1 complex was isolated from *Anabaena* sp. PCC 7120 (hereafter *Anabaena* 7120), and its high-resolution cryo-EM structure (2.22 Å) was presented. The structure's uniqueness lies in an asymmetric ring of multiple IsiA homologs (IsiA1-4) surrounding a monomeric PSI (Figure 1A). In this study, we examine the physiological consequences of removing the belt of antenna proteins (pink and magenta) that comprises the normal IsiA (IsiA3: 4-mer) and the IsiA homolog (IsiA4). We have constructed deletion mutants of *Anabaena* 7120 lacking *all4000* (IsiA4) and *all4001* (IsiA3), individually and in combination. We observed a general increase in oxygen evolution under iron deficiency (Figure 1B-C). This may reflect normalization based on phycobilin levels. Our results also indicate that single deletions of either IsiA3 or IsiA4 positively affect DMBQ-supported oxygen evolution under iron-deficient conditions (Figure 1C). We also performed spectroscopic studies on cells using static absorption and time-resolved fluorescence at 77 K. For each strain, we simultaneously recorded IsiA and PSI fluorescence decay characteristics after preferential excitation of chlorophyll *a*. Subsequently, we performed spectro-kinetic modeling of fluorescence decay maps to evaluate whether structural alteration of the IsiA-PSI supercomplex in the mutated strains induces any changes in IsiA → PSI excitation energy transfer, such as a decrease/increase in the process efficiency.

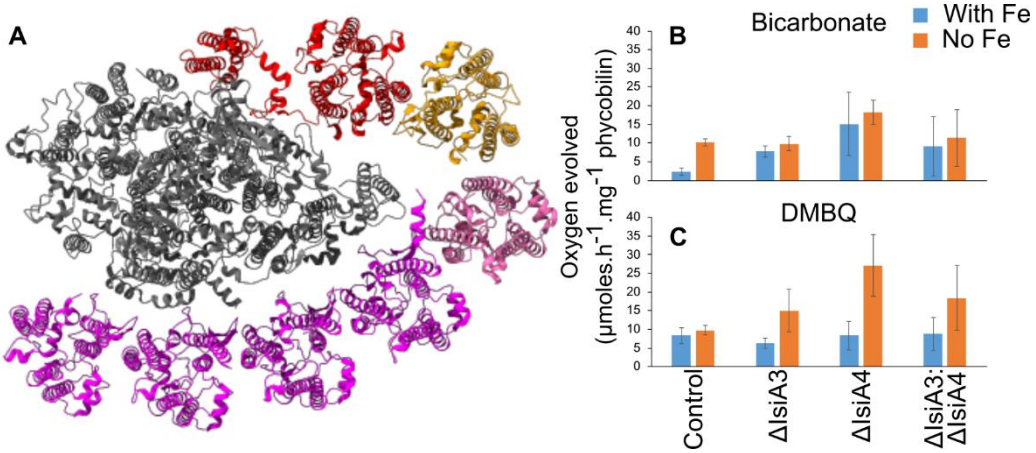


Figure 1. **A**) Unusual IsiA-PSI from *Anabaena* 7120 (PDB; 9ZJT). **B**) Whole-chain oxygen evolution supported by 15 mM sodium bicarbonate. **C**) PSII-specific oxygen evolution supported by 0.2 mM DMBQ and 1 mM $K_3Fe(CN)_6$.

Funding for this study is provided by the U.S. Department of Energy, Office of Basic Energy Sciences, through the grant DE-FG02-99ER20350.

Resolving Phycobilisome Quenching States Through Photon-by-Photon Simulations and Single Molecule Spectroscopy

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In cyanobacteria, the Orange Carotenoid Protein (OCP) and related single-domain homologs prevent oxidative damage by quenching excess excitation energy from the light-harvesting phycobilisome (PBS). Single-molecule spectroscopic techniques such as the Anti-Brownian Electrokinetic (ABEL) trap provide high-resolution data including fluorescence lifetime, brightness, and emission spectrum, which reveal the photophysical states available to phycobilisomes either alone or in complex with other molecules such as photoprotective quenchers. However, inferring structural properties like binding sites and stoichiometry requires pairing this empirical data with accurate computational models of energy transfer. Our group has shown that Monte Carlo simulations using a compartmental model of the phycobilisome faithfully reproduce empirical results to date for known structural models. By simulating the photon-by-photon dynamics of the antenna, we evaluate potential quencher binding sites and combinatorial states across the PBS compartments, including both the allophycocyanin (APC) core and the peripheral phycocyanin (CPC) rods. We test varying configurations and energy transfer rates among compartments or to a quencher, in order to determine which structures and functions are supported by our experimental observations. However, the generalizability of this modeling approach, particularly for variable phycobilisome architecture or for homologs of the quenching proteins, remains unclear. To address this, we systematically evaluate the precision and sensitivity of our simulations to determine the limits of their utility for interpreting multi-parameter single-particle spectroscopic data. We find that the unique structure of the phycobilisome dictates the sensitivity of the simulations to structural perturbations, while precision is largely set by the underlying variability of intact phycobilisome structures. Ultimately, this work establishes the advantages and limitations of using computational predictions to link single-molecule measurements to the physical structure of the light-harvesting complex.

Br- Substitution in Photosystem II of *Chlorella ohadii*

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Photosystem II (PSII) is a large, multisubunit, enzyme complex which catalyzes water oxidation and oxygen evolution. The catalytic site of PSII is the water-oxidizing complex (WOC), which converts water into oxygen, protons, and electrons. There are 2 chloride ions located near the WOC, and these facilitate the removal of protons through specific water channels towards the lumen. Studies involving the substitution of Chloride for Bromide will help better understand the role of Chloride. To date, the role of chloride has been investigated primarily in certain model organisms. The green alga *Chlorella ohadii* was biosynthetically substituted with bromide to investigate chloride function in this clade's PSII. From the 77K spectrofluorometry, the Br-substituted cultures produced less carotenoids than its WT counterpart, suggesting the carotenoids are used for absorption rather than photoprotection under low light. From fast repetition rate fluorometry, the quality of WOC operation for the Br-substituted samples was diminished, which is consistent with similar studies in cyanobacteria. However, the S2 state in the Br-substituted culture seemed to be more prone to reductive decay. From Q_A^- reoxidation, the Br-substituted samples were transferred electrons from the Q_A to Q_B much faster than the WT, while Q_B was also less likely to be in its semi-quinone state. The Oximetry data showed that the Br-grown culture had produced more chlorophyll a and b, but at a similar ratio when compared to the WT. $P700^+$ redox kinetics showed that the substitution did not substantially affect PSI downstream.

Molecular insights into phycobilisome assembly pathway reveal crystalline bodies in cyanobacteria

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Cyanobacteria contain phycobilisome (PBS) complexes that play critical roles in harvesting and directing light energy to the photosystem I and II reaction centers. The proper assembly of PBS components is an intricate process that is required for their activity and association with photosystem I and II. To understand the complex mechanisms regulating the PBS assembly, we knocked out *apcE,* the terminal emitter that also forms the structural scaffold for the PBS core in the cyanobacteria *Picosynechococcus sp.* PCC 7002. *ApcE* knockout led to growth and pigment defects and a decrease in chlorophyll, phycocyanin, and allophycocyanin. Despite decreased chlorophyll, phycocyanin, and allophycocyanin content, the *apcE* knockout displayed increased chlorophyll and phycobilin fluorescence. Fluorescence microscopy revealed the accumulation of highly fluorescent puncta frequently localized to the pole of *apcE* knockout cells. Further investigation using electron cryo-tomography identified large, highly repetitive crystalline arrays of densely packed PBS cylinders in the *apcE* knockout. Together, these data indicate that cyanobacteria may accumulate PBS components in the form of highly organized crystalline bodies as intermediates during PBS assembly.

Investigating the structural basis of photosystem II repair cycle

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Photosystem II (PSII) provides the electrons needed for the reduction of CO_2 to carbohydrate by its light-driven water splitting reaction. This demanding redox reaction inevitably produces reactive oxygen species that damage PSII. Understanding how plants repair this damage can inform strategies to improve photosynthetic efficiency and crop yield. Damaged plant PSII has been known to undergo a dynamic disassembly and repair cycle in which the PSII supercomplex dissociates into smaller complexes so that the damaged subunits can be removed and replaced with de novo synthesized proteins. Our recent work suggests that PSII phosphorylation plays a role in this process by aiding in the disassembly of the supercomplexes and preventing PSII subunits from reassembling prematurely. Our understanding of the PSII repair cycle has, however, been limited by lack of information on the composition and structure of the final disassembly intermediates. Study of these disassembly intermediates is currently limited by their low abundance. Our lab has created a double mutant that combines the overexpression of the core kinase STN8 (STN8oe) with a loss of function of the core phosphatase PBCP (*pbcp-1*). This double mutant has a higher level of PSII phosphorylation, which in turn leads to increased accumulation of the repair intermediates. This makes it an ideal source for isolating the disassembly species for structural characterization. Purified disassembly intermediates are currently being characterized via cryo-electron microscopy (cryo-EM). We have obtained structures of monomeric cores (C1) and reaction center 47 (RC47). Our structural data reveal preliminary protein and pigment composition of these complexes. Cryo-EM studies of the disassembly intermediates have the potential to shed light on the structural nature of the photodamage and the identity of any assembly/repair factors that associate with C1 or RC47, providing new directions for studies of PSII repair and assembly mechanisms.

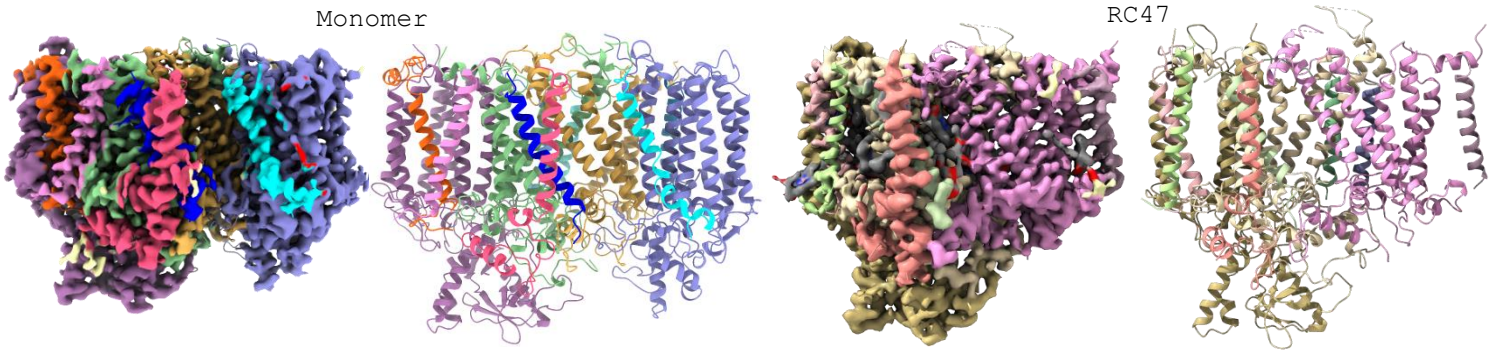


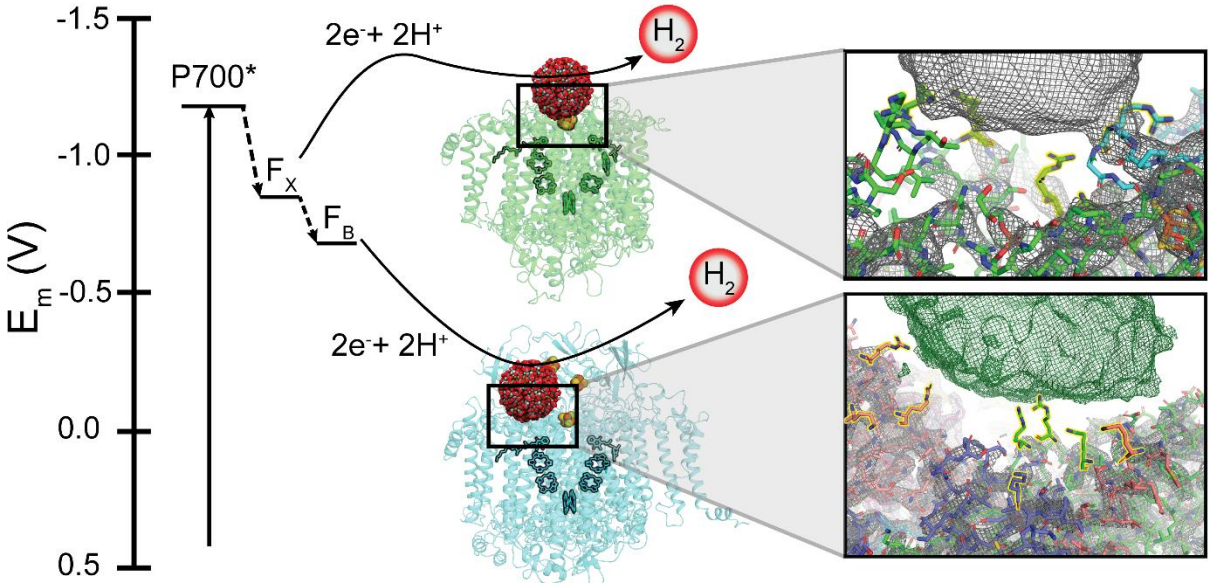
Figure. 1. Ab-initio reconstruction of monomeric core and RC47 Structures. Cryo-EM density and ribbon diagram for each complex are shown side-by-side.

Molecular Design Principles for Photosystem I-Based Biohybrid Solar Fuel Catalysts

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Photosynthesis relies on the solar-to-chemical energy conversion activity of the photooxidoreductases Photosystem I (PSI) and Photosystem II (PSII). Directing the resulting chemical energy toward the production of fuels, this activity can be leveraged to generate clean, dispatchable fuels from abundant resources using solar energy. One promising method uses platinum nanoparticles (PtNPs) that are engineered to bind the surface of Photosystem I at the native electron-acceptor binding site, allowing the PtNPs to harvest photoexcited electrons from F_B, the terminal electron transfer cofactor of PSI. In a variation on this system, the same PtNPs readily bind a chemically modified PSI which has had the stromal subunits removed, providing access to F_X from which higher-energy electrons may be harvested. Here we present three PSI-PtNP biohybrid structures as well as molecular dynamics simulations describing the dynamic interactions between the protein and nanoparticles. These data show that functional inefficiencies are related to both intrinsic electron transfer limitations and the distribution of binding geometries allowed by the protein architecture. Drawing design principles from these, we are now taking alternative approaches to biohybrid assembly to diversify catalyst compatibility and designing new PtNP-binding domains for highly controlled PtNP binding and incorporation into alternative protein-based photocatalysts.



Optical Spectra Prediction of pH-dependent Wild-type and Site-Mutated Water-Soluble Chlorophyll Protein of Lepidium Virginicum (LvWSCP)

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Water-Soluble Chlorophyll Proteins (WSCP) are known for their stability and resistance to photodegradation of the bound chlorophylls. These systems have a relatively simple structure with well-defined, symmetric chlorophyll-binding sites. Although the structural features of chlorophyll-containing proteins are well established, accurately predicting their optical and electronic properties remains challenging because modeling large, electronically coupled pigments within a dynamically fluctuating protein environment is complex. Another factor to account for in molecular modeling is the possible effect of pH-driven changes in protein amino acid protonation states, which may perturb local electrostatics and pigment-protein interactions, thereby influencing electronic excitation energies and spectral frequency shifts.

This work focuses on elucidating how pH-dependent changes in the protein environment govern the optical properties of the chlorophyll pigments in both wild-type and mutated WSCP. Our methodology integrates molecular dynamics (MD) simulations with quantum mechanics/molecular mechanics (QM/MM) constrained geometry optimizations, followed by polarizable excited-state QM/EFP (EFP: Effective Fragment Potential) calculations. Additionally, QM/EFP-based Pairwise Excitation Energy Decomposition Analysis (PEEDA) assesses molecular-level details of how the protein environment influences excitation energies. By identifying the mechanistic link between pH, protonation states, and spectral tuning, this work offers insights into how the protein environment controls spectral shifts and shapes energy transfer pathways between pigments, informing the rational design of photosynthetic systems with tailored optical properties.

Stomatal engineering for improved water use efficiency: an automated workflow

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Stomatal responses to changing light influence the balance between photosynthetic carbon gain and water loss, making guard cell signalling an attractive target for improving plant water use. However, evaluating candidate regulators at scale remains constrained by the time and effort required to generate edited plants and characterise their phenotypes. Here, we present an improved pipeline for high throughput plant engineering and automated phenotyping in *Nicotiana benthamiana*, designed to connect candidate genes with stomatal structure and function. The workflow combines streamlined generation of edited lines with machine learning assisted stomatal and morphometric analysis, alongside dynamic thermal and gas exchange phenotyping. Using BLUS1, a protein kinase linking blue light perception to stomatal opening, we examined how targeted genetic changes influence stomatal responses to light. Across four BLUS1 targeted T2 lineages, increasing in stomatal conductance following a red to red–blue illumination transition increased by only 9–11% , compared with 27% in control plants. This attenuated response was accompanied by an increase in intrinsic water use efficiency during the spectral transition, highlighting BLUS1 signalling as a candidate for modifying the balance between carbon uptake and water loss. Together, these findings illustrate how integrated plant engineering and deep learning approaches can resolve stomatal function and support the evaluation of strategies to improve plant water economy.

Probing the effect of chemical environment on α-carboxysomes shell stability

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Carboxysomes (CBs) are self-assembling proteinaceous bacterial microcompartments part of the Calvin-Benson cycle that convert inorganic carbon dioxide (CO_2) into organic carbon as 3-phosphoglyceric acid. While there is interest in expressing CBs in plants (for example, to increase crop yields), the redox environment, salinity, and pH in a plant chloroplast is distinct from those of the native CB host, *Halothiobacillus Neapolitanus*. Chemical environment has been observed to modulate subunit-subunit interactions in simplified systems, but its impact on shell stability has yet to be explored on whole CBs. This study probes the stability of intact CBs in varying pH, salinity, and redox conditions. Single molecule total internal reflection fluorescence (smTIRF) microscopy will be used to observe the exchange between destabilized CB shells and fluorescently labeled exogenous shell subunits. Simultaneous imaging in two detection channels will allow fluorescent monitoring of originally incorporated shell subunits and exogenous subunits under these various conditions. These results will ascertain the stability of the CB in various chemical environments, identifying limitations to overcome as they are incorporated into new hosts.

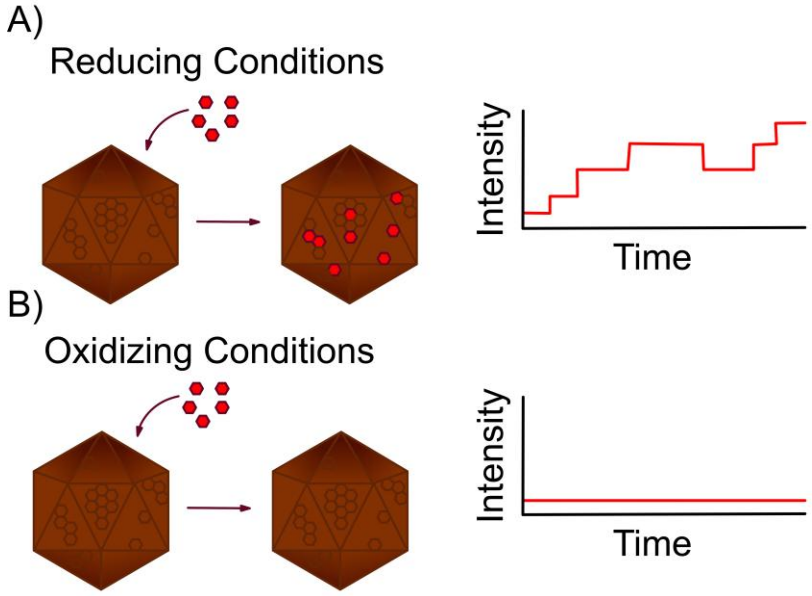


Figure. 1. Carboxysome association model for labile and static Conditions. A) Reducing conditions depicted here are examples of labile conditions where red labeled exogenous shell subunits are associated to orange labeled carboxysomes. The addition of shells is shown in a stepwise fashion to the right. B) Oxidizing conditions as an example to show little to no association. This is indicated as the intensity over time is constant, shown to the right.

Photosynthetic responses to dynamic light within the maize canopy

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Light energy powers photosynthesis and in turn supports crop growth and productivity, but light is rarely constant in the open fields where most crops are grown. Rapid fluctuations in irradiance arise from cloud cover, wind, sun position, and self-shading, requiring leaves to continually adjust photosynthetic activity to changing light availability. These dynamics are particularly complex within crop canopies, where canopy structures create substantial variation in both average irradiance and the magnitude and frequency of light fluctuations. Models of canopy photosynthesis show how these differences substantially influence whole-canopy carbon assimilation, but open questions remain regarding the specific responses to fluctuations in light across the canopy. In this study, we investigated differences in carbon assimilation under dynamic light conditions at two canopy levels (middle and top) in field-grown maize, during both vegetative and reproductive stages. To simulate the varying light conditions that top and middle leaves experience, two dynamic light protocols were utilized to measure carbon assimilation: simulating sudden shade during a sunny day, and sunflecks in the shaded canopy on both top and middle leaves. We found that top leaves had generally higher assimilation than middle leaves—both activating photosynthesis more rapidly in response to sudden increases in light, as well as sustaining higher assimilation during decreases in light. However, this advantage diminished with larger fluctuations in irradiance and was lost under strongly limiting light. These results show that photosynthetic responses to dynamic light vary with canopy position and with the direction and magnitude of changes in irradiance, and that leaves are not uniformly optimized for the dynamic light environments experienced across the canopy. This identifies the kinetics of the lower canopy light response as a target for engineer efforts to maximize whole-canopy carbon assimilation.

Decoding C_4 Photosynthesis to Develop Smart Sorghum with Enhanced Photosynthetic Efficiency in the Field

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Sorghum bicolor is a leading candidate bioenergy feedstock, but its productivity is frequently limited by challenging environmental conditions in the field, e.g. excess sunlight, heat, drought and others. Under field stresses, photosynthesis is compromised and a normal light becomes excessive, resulting in damaged photosynthetic machinery and accumulated reactive oxygen species (ROS). Although the top leaf canopies often experience excess light, the dense plantation in the sorghum field can also result in light shading at the bottom of the canopy. Moreover, sorghum uses two coordinated cell types for C4 photosynthesis: mesophyll (M) and bundle sheath (BS) cells. It is under investigation how C4 photosynthesis responds to abiotic stresses in the real-world field environments at different time points and different leaf developmental stages. We employed field-grown sorghum plants to investigate the limitation of C_4 photosynthesis at different developmental stages and from dawn to dusk. Using a LI-COR LI-6800, we measured photosynthetic and gas exchange parameters across three leaf developmental stages (top young, top mature, bottom mature) at three time points (9 AM, 1 PM, 5 PM). Chlorophyll fluorescence, gas exchange, light-response curves at (0–2000 µmol m^{-2} s^{-1}), and CO_2 response (A–Ci) curves were used to assess photosynthetic performance in leaves at different developmental stages at different time points. Our results showed the differences in photosynthetic efficiencies and redox status increased over time among the three leaf developmental stages, suggesting temporal dynamics of photosynthetic performances in leaves at different canopy levels and developmental stages. These physiological data in sorghum leaves will be integrated with 3D volume imaging, omics analysis, predicted regulatory networks, and functional validation to guide the engineering of climate-resilient and smart sorghum with improved photosynthesis for sustainable bioenergy production.

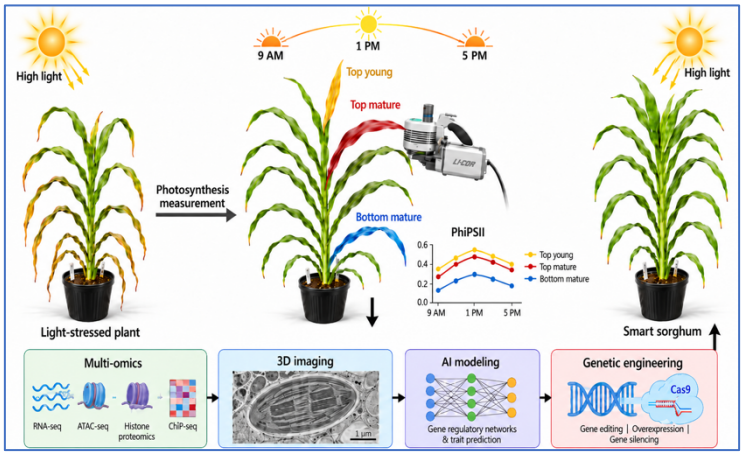


Figure: Illustration showing the experimental plan to dissect the photosynthetic performance of field grown sorghum in different light and temperature conditions of the day and future strategies to engineer smart sorghum which has better photosynthetic efficiency.

Charge-Transfer States of the Reaction Center Complex of Photosystem I: Insights from Electronic Structure Calculations

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The initial charge separation event in Photosystem I (PSI) relies on the reaction center (RC) composed of six strongly interacting Chlorophyll a (Chl) molecules. To understand charge-separation dynamics within this complex, we are constructing the RC electronic Hamiltonian from first principles. The site energies for the Hamiltonian are obtained from the polarizable embedded QM/EFP (hybrid quantum mechanics / effective fragment potential) computations of the whole PSI complex. These calculations show that the long-range CT states (special pair to A0) are sufficiently low in energy to be directly accessible as primary charge-separation states. In this work, we mainly concentrate on computing electronic couplings between locally excited states and charge-transfer (CT) states across Chl pairs. The couplings were quantified utilizing fragment-charge difference (FCD) and fragment-excitation difference (FED) methods, at the ωB97X-V/6-311G* level of theory. To probe the sensitivity of these electronic couplings to spatial fluctuations, we systematically calculated the couplings for the special pair dimer under horizontal and vertical geometric displacements. Ultimately, this comprehensive mapping of the coupled excited and CT states provides deeper physical insight into the process of the initial photosynthetic charge separation event in the PSI reaction center.

Monitoring photosynthesis in the field: Development of a plant-wearable fluorometer

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Photosynthesis dynamically responds to environmental changes across multiple timescales, from second-scale NPQ to long-term acclimation. We're highly interested in how plants apply different strategies to keep the best photosynthetic efficiency while being protected from dramatic light fluctuations. Traditional outdoor survey measurements can only take snapshots without the context of what the plants have experienced. Environment-controlled chamber experiments can track both environmental factors and plant responses, but indoor setups cannot fully simulate outdoor conditions (*e.g.*, solar spectrum, incident angle, wind, *etc*.).

Here, we want to present our recent development of the "Ambit" sensor, a scalable plant-wearable fluorometer that can monitor fluorescence parameters *in situ* for months. Ambit is a low-cost ($50), field-tested (−20 °C in Michigan to 60 °C in the Australian Outback), miniature (see picture), and open-source PAM-like fluorometer designed for long-term monitoring of variable fluorescence induced by ambient light or added light. Paired with Ambyte (a data logger and power bank), Ambits can record photosynthesis for months in the field without human intervention.

Along with the physical devices, we also want to present what we've learned from recent field deployments, including plant responses to dynamic light fluctuations, canopy structures, and temperatures.



Figure. Ambit sensors under deployment. (A) Ambytes (each connect to 4 Ambits) in a wheat field. (B) An Ambit applying saturation flashes. (C) Example of Ambit data. Green: light intensity, red dots: temperature; blue (F_S) and red (F_M`) vertical lines: chlorophyll fluorescence.

Phytohormone-associated Interactions between Antarctic Bacteria and Algae

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It is highly probable that interactions between bacteria and phytoplankton have a large influence on the fixation and flow of carbon in most aquatic ecosystems on our planet. Many of these interactions take place within the phycosphere, a microecosystem around the periphery of individual phytoplankton cells. Within the phycosphere, biologically-derived molecules, including organic carbon compounds and nutrients are actively excreted and exchanged. Recent studies have found that the phycosphere is also associated with signaling between these microbial partners, including phytohormones which may be algal- or bacterial-derived. There appears to be several complex outcomes of phytohormones on algal growth and physiology, including increased excretion of organic carbon, larger cell sizes, and reductions in cell proliferation rates. Phytoplankton cells can also gain improved resistance to abiotic stress. We studied algal-bacteria interactions using isolates from perennially ice-covered lakes located in the McMurdo dry valleys, Antarctica. These lakes are dominated by a microbial food web and receive minimal external inputs, resulting in tight reliance on algal production of organic matter. While this ecosystem has been studied for more than 3 decades, the existence of bacteria-algal interactions or a phycosphere is unknown. We hypothesized that owing to the strong reliance on new algal-derived carbon production combined with the permanent abiotic stress which life is exposed to, there is a phycosphere in ice-covered Antarctic lakes which may be influenced by bacteria-derived phytohormones. To test this prediction, we [1] developed algae-bacterial co-cultures from lake water samples collected from Antarctic lakes (McMurdo Dry Valleys, Antarctica); [2] isolated and identified a number of bacteria from the co-cultures, [3] performed genomic sequencing on 10 strains of bacteria and queried the data for putative phycosphere and phytohormone synthesis pathways, [4] quantified production of a prevalent phytohormone, indole-3-acetic acid (IAA). Many of the bacterial isolates contain partial pathways for the production of IAA, within their genomes. The isolates also contain other pathways commonly found in phycosphere-associated bacteria, such as chemotaxis, high transporter count, and attachment. Last, many of the bacteria also exhibited IAA production in vivo. The implications of these findings for Antarctic algal growth and physiology will be discussed.

Growth Response of *Cyanothece* 51142 Under Differential Guanidine Concentration

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Cyanothece sp. ATCC 51142 is a unicellular cyanobacterium that has the ability to perform oxygenic photosynthesis and fix nitrogen into bioavailable ammonia. Because guanidine becomes toxic at elevated concentrations, this experiment examined the growth and tolerance of *Cyanothece* 51142 under increasing guanidine concentrations in ASP_2+N and ASP_2-N mediums. We hypothesized that through long-term cultivation, *Cyanothece* 51142 will be able to adapt to higher levels of guanidine. Initially, *Cyanothece* cultures were analyzed in guanidine concentrations ranging from 0.0 to 3.0 mM to establish a toxicity baseline. We then used adaptive laboratory evolution (ALE) with long-term cultivation conducted in multi-cultivator photobioreactors. Cultures were systematically exposed to increasing levels of guanidine over time. Light microscopy was used to examine the cultures throughout the growth experiment. We were able to isolate two evolved *Cyanothece* 51142 strains that were able to grow in guanidine concentrations up to 5-fold higher than WT. These evolved *Cyanothece* 51142 strains grown under nitrogen replete conditions were sent for whole genome sequencing. The sequencing results identified multiple DNA changes in evolved cultures, including insertions, deletions, and base pair changes. These results demonstrate that *Cyanothece* 51142 can adapt to guanidine concentrations that initially inhibited wild-type growth and suggest that genetic changes may contribute to increased guanidine tolerance. We will continue to analyze the evolved genomes and perform additional sequencing that will help pinpoint genetic changes associated with adaptation. Future experiments will also determine whether ASP_2-N cultures can similarly evolve to tolerate higher guanidine concentrations. This work was supported by the U.S. Department of Energy, SFEE program (DE-SC002567) and RENEW program (DE-SC0024702).

The role of thylakoid galactolipids in LCNP- and ROQH1-dependent regulation of qH non-photochemical quenching studied in a proteoliposome system

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Non-photochemical quenching (NPQ) is an important photoprotective mechanism that dissipates excess absorbed light energy as heat, thereby limiting the formation of reactive oxygen species (ROS) and protecting the photosynthetic apparatus from oxidative damage. Several components of NPQ have been characterized, including the rapidly reversible qE, the more slowly relaxing or sustained components qI, and qH. Among these, qH is particularly intriguing because it persists long after transfer to darkness, does not depend on lumen acidification or PsbS, and relies on a molecular mechanism distinct from that of the well-characterized qE pathway.

Genetic and biochemical studies have identified two thylakoid-associated proteins involved in qH regulation: the lumenal plastid lipocalin LCNP, which is required for qH induction, and ROQH1, an atypical member of the short-chain dehydrogenase/reductase (SDR) superfamily involved in qH relaxation. Accordingly, plants lacking LCNP are unable to establish qH, whereas *roqh1* mutants display slower recovery from sustained quenching. However, the molecular mechanisms through which these proteins regulate qH remain unclear.

Thylakoid membranes are enriched in the non-bilayer-forming galactolipid monogalactosyldiacylglycerol (MGDG), which influences membrane curvature and other biophysical properties. Given preliminary data which found that recombinant LCNP binds to MGDG, and the involvement of SDR enzymes in lipid-related metabolism, we hypothesize that LCNP and/or ROQH1 acts on MGDG and thereby influence qH-related energy dissipation.

To investigate this hypothesis, we use a thylakoid-inspired proteoliposome system in which light-harvesting complex II (LHCII) is reconstituted into large unilamellar vesicles containing a thylakoid-inspired lipid mixture, with or without MGDG. Purified recombinant LCNP and ROQH1 are added externally, individually and in combination, to assess their effects on LHCII-associated chlorophyll fluorescence and energy dissipation.

By systematically varying protein and lipid composition, this approach will allow us to determine whether MGDG modulates LHCII-associated energy dissipation in the presence of LCNP and ROQH1 and whether any observed effects arise from changes in membrane biophysical properties, direct protein–lipid interactions, or both. This study aims to provide mechanistic insight into qH and advance our understanding of how the thylakoid lipid environment contributes to the regulation of photoprotection in plants.

Unraveling the Role of the Lutein Epoxide (LxL) Cycle: A Field Evaluation of LEP9 CRISPR-Cas9 Soybean Mutants

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Beside the well-studied β-xanthophyll (VAZ) cycle, the lutein epoxide (LxL) cycle regulates absorbed excitation energy during photosynthesis in response to light. Not ubiquitous in the plant kingdom, the LxL cycle functions parallel to the VAZ cycle, operating primarily in shaded canopies in most systems studied to date. It involves the interconversion of the α-xanthophyll lutein epoxide (Lx) into lutein (L) under high light by violaxanthin de-epoxidase (VDE), while under low light, L is slowly epoxidized back to Lx by lutein epoxidase (LEP). This field experiment evaluated the phenotype of CRISPR-Cas9 mediated *LEP* knock-outs (KO) in soybean to disentangle the VAZ and LxL cycles and clarify the role of the α-xanthophyll cycle in photosynthesis. Plants were grown during the 2026 field season in a randomized block design with four blocks. Alongside with wild type plants (WT - LD11), three KO lines were evaluated during vegetative and reproductive stages for changes in carbon assimilation and NPQ response. Step-change light routines utilizing an infrared gas analyzer and a chlorophyll fluorescence imager tested responses to high and low light in both shaded (bottom) and sun-exposed (top) leaves. During vegetative low-to-high step-changes, KO lines showed no statistical differences compared to WT, despite KO lines tended to have slightly higher carbon assimilation across all leaves, whereas during the reproductive stage, the trend shifted towards similar or lower assimilation in KO lines relative to WT. During downward light steps, there was no observable difference in assimilation between lines regardless of developmental stage. Non-photochemical quenching induction and relaxation did not differ between lines at any stage, indicating that the absence of Lx did not influence NPQ kinetics in stable lines. However, significantly decreasing Lx content across stages and leaf positions led to increases in L content, which ultimately increased accumulation of the α- xanthophyll pool relative to WT. Interestingly, β-xanthophylls were also variably affected, showing an overall increase on the β-branch observed at the pod stage. Overall, disrupting the LxL cycle alone seems to not impair leaf carbon assimilation or photochemistry, but may trigger compensatory shifts in β-xanthophylls. Further analysis will be needed to clarify the mechanisms driving this adaptation.

This work was supported by Gates Agricultural Innovations grant investment ID 57248.

Low pH State Effect is Real

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Low pH can induce State Transitions in photosynthesis, which affects the way that light is distributed between Photosystem I (PSI) and Photosystem II (PSII). This pH-induced state transition can be observed by 77 K chlorophyl florescence but has not previously been confirmed by biochemical techniques like Clear Native PAGE. State Transitions (ST) occur when the light-harvesting complexes of PSII (LHCII) are redistributed from PSII to PSI, ostensibly to balance the energy absorbance between the photosystems. State Transitions are regulated by the Stn7 kinase, which is in turn activated by reduction of the plastoquinone pool. Phosphorylation of LHCII by Stn7 allows it to move from PSII (state 1) to PSI (state 2), which is observable as an increase in PSI chlorophyll fluorescence. Non-Photochemical Quenching (NPQ) is a photoprotective mechanism that disperses excess light as heat to prevent damage PSII. Although low pH is known to regulate NPQ through protonation of the PsbS protein, the mechanism by which low pH might induce a state transition is not known. We investigated the effects of pH on thylakoid organization by Clear Native PAGE to identify changes in PSI-PSII complexes and PSI-LHCII super complexes. The treated thylakoids and separated bands were then analyzed using 77 K chlorophyll fluorescence to examine changes in energy distribution between PSI and PSII. The results showed that exposure to low pH promoted the formation of PSI-PSII complexes and PSI-LHCII super complexes, which was correlated with State Transitions.

Spectroscopic signatures of low-energy states in chlorophyll *d*-based Photosystem I

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The cyanobacterium *Acaryochloris marina* performs oxygenic photosynthesis using chlorophyll *d*, a red-shifted tetrapyrrole that enables light harvesting beyond 700 nm. Although strains exhibit distinct spectral phenotypes, the origin of this variability has remained unclear. Here, we show that differences in far-red absorption arise exclusively from Photosystem I (PSI) by isolating pigment-protein complexes and comparing their spectroscopic properties across representative short-, intermediate-, and long-wavelength strains.

Low-temperature absorption and time-resolved fluorescence reveal multiple, strain-dependent low-energy chlorophyll states within PSI. Short-wavelength PSI is dominated by an ~730 nm emitter, whereas intermediate and long-wavelength variants exhibit additional states near ~745 nm and ~760 nm with distinct kinetic signatures. Transient absorption measurements resolve ultrafast excitation flow from higher-energy chlorophylls into these low-energy states and the reaction center, demonstrating that excitation trapping pathways are altered but remain efficient across spectral types.

Spectroscopic differences are interpreted in the context of high-resolution cryo-EM structures of PSI from intermediate strains, which reveal subtle variation in chlorophyll environments within a conserved scaffold. Phylogenetic analysis further supports a model in which tuning of chlorophyll excited-state energies has emerged through multiple evolutionary routes rather than a single conserved structural solution. Together, the data indicate that similar low-energy states can arise from distinct pigment-protein interactions, highlighting the intrinsic tunability of chlorophyll tetrapyrroles within PSI.

This work establishes PSI as the molecular origin of spectral diversity in *A. marina* and provides a framework for understanding how tetrapyrrole-protein interactions extend light harvesting into the far-red.

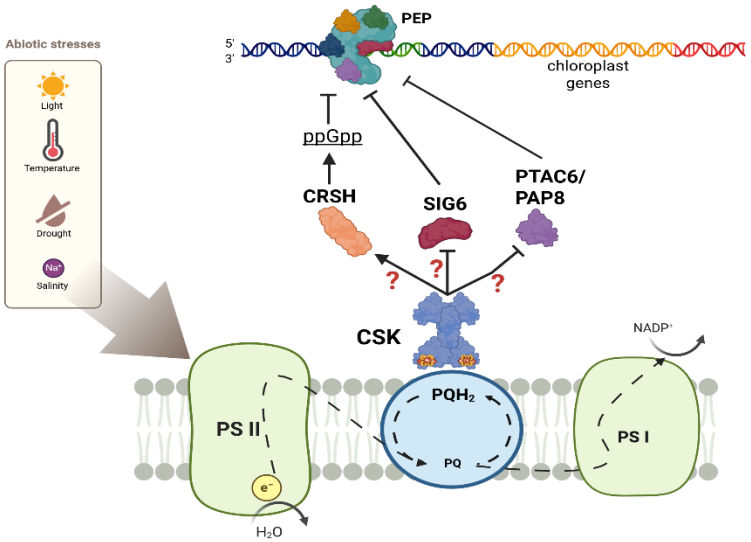
Identifying the Functional Partner(s) of Chloroplast Sensor Kinase in Land Plants

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Chloroplast Sensor Kinase (CSK) is an evolutionarily conserved bacterial-type kinase present in photosynthetic organisms. In cyanobacteria, the CSK homologue, Hik2, functions within a canonical two-component signaling system with its response regulator Rre1 to place transcription of certain heat stress genes under the redox control of plastoquinone (PQ). In contrast, CSK in plants has lost both the conserved histidine autophosphorylation site and its cognate response regulator partner. Nevertheless, CSK in these lineages retains the capacity to bind a 3Fe-4S cluster essential for redox sensing of the PQ pool. Previous studies have shown that lack of CSK alters chloroplast gene expression, with our recent unpublished research indicating that acute high intensity light exacerbating these changes. Additionally, CSK mutants show a growth phenotype under prolonged salinity and osmotic stress, suggesting that CSK may mediate tolerance to multiple abiotic stresses. To map the complete CSK signaling pathway from thylakoid to-chloroplast gene expression machinery, we used a combination of targeted yeast-two hybrid assays and TurboID-based proximity labelling. Our data shows CSK interaction with chloroplast sigma factor 6 (SIG6), as well as a strong proximity of CSK to Calcium-activated RelA/SpoT Homolog (CRSH) and PLASTID TRANSCRIPTIONALLY ACTIVE protein 6 (pTAC6). Each of these three proteins are associated with the chloroplast transcriptional machinery and may be targets of CSK phosphorylation, adjusting transcriptional activity in response to PQ reducing conditions. In light of this interactome data, we hypothesize that CSK links photosynthetic redox reactions with chloroplast transcription by direct interaction with the proteins of the chloroplast transcriptional machinery. Validation of these candidate CSK partners and characterizing the nature of their interactions, will be essential in defining this novel regulatory pathway in the chloroplast.

Figure 1. Hypothetical model of CSK-dependent transcriptional regulation in the chloroplast in response to abiotic stress.



Are the Master Regulators of CA4 AraC Family Proteins? The Search for CA4 Photoreceptors

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Marine *Synechococcus* cyanobacteria exhibit remarkable pigment diversity and dynamically adjust their light-harvesting antennae in response to changes in the ambient blue-to-green light ratio[1]. This process, known as Type 4 Chromatic Acclimation (CA4), is widespread among marine *Synechococcus*, occurring in more than 40% of surveyed populations, and is associated with acclimation to low-light conditions[2]. All of the genes required to impart the CA4 response are encoded within small genomic islands in CA4-capable marine *Synechococcus* isolates. Two of these genes are the regulatory proteins FciA and FciB. Previous studies have established these proteins as central regulators of CA4[3]. However, the molecular mechanism by which they perceive changes in light remains unknown.

FciA and FciB are particularly intriguing because both belong to the AraC family of transcriptional regulators[4], a protein family not previously known to function as photoreceptors. Their proposed role in light sensing therefore raises the possibility of a previously unrecognized class of photoreceptors and a novel mechanism of environmental light perception. To investigate this hypothesis, we generated genes encoding tagged versions of FciA and FciB and inserted these into a self-replicating plasmid, then introduced these constructs into corresponding *Synechococcus fciA* and *fciB* mutant strains and tested for complementation of the mutant phenotypes. The complemented strains are providing a system for isolating tagged versions FciA and FciB from their native cellular environment and testing these proteins in order to determine whether these proteins are associating with light-responsive chromophores. We are purifying these tagged proteins and characterize their associated pigments using biochemical and spectroscopic approaches. Identification and characterization of a chromophore associated with FciA or FciB would provide critical evidence for their proposed role as photoreceptors and advance our understanding of the molecular mechanism underlying CA4 potentially revealing a new class of cyanobacterial photoreceptors.

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[1] Joseph E. Sanfilippo et al., "Chromatic Acclimation in Cyanobacteria: A Diverse and Widespread Process for Optimizing Photosynthesis," *Annual Review of Microbiology* 73, no. 1 (2019): 407–33, https://doi.org/10.1146/annurev-micro-020518-115738.

[2] Théophile Grébert et al., "Light Color Acclimation Is a Key Process in the Global Ocean Distribution of *Synechococcus Cyanobacteria*," *Proceedings of the National Academy of Sciences* 115, no. 9 (2018), https://doi.org/10.1073/pnas.1717069115.

[3] Joseph E. Sanfilippo et al., "Self-Regulating Genomic Island Encoding Tandem Regulators Confers Chromatic Acclimation to Marine *Synechococcus*," *Proceedings of the National Academy of Sciences* 113, no. 21 (2016): 6077–82, https://doi.org/10.1073/pnas.1600625113.

[4] Daniel Cortés-Avalos et al., "An Update of the Unceasingly Growing and Diverse AraC/XylS Family of Transcriptional Activators," *FEMS Microbiology Reviews* 45, no. 5 (2021): fuab020, https://doi.org/10.1093/femsre/fuab020.

Mutations Near the Chlorophyll Pocket Drive Changes in Dipole Orientation in WSCP

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The green plant water-soluble chlorophyll (Chl) protein (WSCP) provides a well-defined model system for investigating how the local protein environment influences chlorophyll excitonic interactions and optical properties. Its highly symmetric homotetrameric architecture, structurally characterized pigment-binding pocket, and tolerance to mutagenesis make WSCP an attractive platform for probing structure-function relationships in chlorophyll-protein complexes. While previous studies have primarily emphasized electrostatic tuning of chlorophyll site energies, the contributions of other factors, including steric interactions, hydrogen-bonding rearrangements, and changes in pigment geometry, remain less fully characterized.

In this work, we explore structural tuning in a library of 33 single and double mutants targeting residues surrounding the Chl macrocycle and phytol-tail region, including V35, T52, S53, L54, and Q57 variants. Not all mutants successfully reconstituted with Chl, providing insight into binding-site constraints and protein-pigment compatibility. Variants that reconstituted were characterized using room-temperature UV-visible absorption and circular dichroism (CD) spectroscopy, together with 77 K absorption and fluorescence spectroscopy, to evaluate pigment incorporation, excitonic structure, and excited-state behavior. Most substitutions produced modest spectral perturbations or reduced reconstitution efficiency, particularly among charged residue substitutions. However, multiple mutants exhibited sufficiently strong pigment incorporation to allow direct comparison of their excitonic features with those of wild-type WSCP. The most pronounced spectroscopic changes were observed for mutations at Val35. In particular, V35Y and V35W displayed substantial redistribution of oscillator strength between the upper and lower Qy exciton bands. The V35Y mutant additionally exhibited enhanced CD intensity and a narrowed 77 K fluorescence spectrum, consistent with altered excitonic interactions and excited-state delocalization. Excitonic simulations indicate that the observed changes are not readily explained by shifts in site energies alone. Rather, the results suggest that mutation-induced modifications to chlorophyll orientation and transition-dipole geometry contribute significantly to the observed spectral behavior. These findings further illustrate how targeted protein engineering can modulate excitonic properties in WSCP and provide useful design principles for the development of biomimetic light-harvesting systems and photonics.

Investigating Carbon-Concentrating Bacterial Microcompartment Assembly

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Carboxysomes are bacterial microcompartments that serve as sites of carbon fixation and are composed of a proteinaceous outer shell, carbonic anhydrase, and RuBisCO, the enzyme responsible for catalyzing the first step of the Calvin cycle. Oxygen directly competes with carbon dioxide for the same RuBisCO active site, making the carboxylation reaction slow and inefficient. To counteract this inefficiency, some cyanobacteria and chemoautotrophs, such as *Halothiobacillus neapolitanus,* have evolved carbon-concentrating carboxysomes that increase the local concentration of bicarbonate within a proteinaceous shell and allow more efficient carboxylation reactions. When carbon dioxide availability in the environment is low, α-carboxysomes assemble within *H. neapolitanus* cells and facilitate the enzyme-catalyzed conversion of inorganic carbon to organic carbon. RuBisCO molecules polymerize to form a lattice near the shell of the carboxysome, but the mechanism by which RuBisCO molecules assemble within the microcompartment remains unknown. Immunoblotting techniques are being used to identify how ratios of α-carboxysome scaffold protein isoforms, CsoS2A and CsoS2B, and RuBisCO structural forms compare in high and low carbon dioxide availability environments. Cryo-electron tomography is used to visualize ultrastructures within *H. neapolitanus* and provide insight into how carboxysomes and RuBisCO are situated within the cell. The goal of this project is to reliably develop a pipeline to induce microcompartment assembly and visualize stages of maturation using cryo-EM and cryo-ET. This will allow for 3D reconstructions of α-carboxysomes with varying assembly morphologies in order to deduce how RuBisCO is packaged within carboxysomes. Understanding microcompartment assembly and carbon-concentrating mechanisms used by bacteria will open up avenues for biotechnologists to develop nanoproducts that improve photosynthetic efficiency and crop yield.

Optimization of Exoelectrogenic Activity in the Photosynthetic Bacterium *Rhodobacter capsulatus* for Bioelectricity Generation

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Photosynthetic bacteria possess versatile metabolic pathways that can support extracellular electron transfer and provide opportunities for sustainable bioelectricity generation. In this study, a low-cost photosynthetic microbial fuel cell (PMFC) system was developed to redirect electrons derived from cellular metabolism toward electricity generation rather than hydrogen production. *Rhodobacter capsulatus* was used as the anodic microorganism in a double-chamber PMFC connected by a potassium chloride (KCl) salt bridge. Key operational parameters, including inoculum volume, pH, and catholyte salt concentration, were optimized using response surface methodology (RSM).

Bioelectrogenic performance was evaluated based on voltage and current generation, while optical density at 660 nm (OD_{660}) was monitored as an indicator of microbial growth and physiological response. The optimized conditions consisted of a catholyte concentration of 6 mM, an inoculum volume of 150 mL, and pH 7. Under these conditions, the system produced a maximum voltage of 1.38 V and a current of 0.15 A, with a corresponding OD_{660} of 0.837. These results demonstrate that optimization of physicochemical and biological parameters can enhance the bioelectrogenic performance of *R. capsulatus* in a photosynthetic microbial fuel cell.

The findings highlight the potential of photosynthetic microorganisms as biological platforms for converting metabolic reducing power into electrical energy. Optimization of electron-transfer conditions in *R. capsulatus* may contribute to the development of low-cost bioelectrochemical systems for simultaneous bioenergy recovery and wastewater treatment.

Key words: Photosynthetic Microbial Fuel Cell, Bioelectricity, bioelectrochemical,

MazF-Mediated CRISPR: An Improved Approach for Marine Cyanobacterial Transformation

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Cyanobacteria are photosynthetic organisms that have critical roles in maintaining environmental and agricultural balance. Moreover, the photoautotrophic nature of these organisms is establishing them as promising bioengineered factories for producing sugars, biofuels, vitamins and cosmetics[1]. Recent developments in genetic engineering tools such as CRISPR/Cas12a has facilitated genetic transformation of cyanobacteria, however, it is still a time-consuming process that makes it challenging to make multiple modifications in a single strain [2]. Several counter-selection systems such as *sacB* and *acsA* have been developed but are not efficient for transformation in all cyanobacterial strains. Recently, *mazF*, a gene encoding an endoribonuclease toxin protein in *Escherichia coli,* was evaluated as a novel counter-selection marker for several freshwater cyanobacterial species and showed encouraging results [3]. In our study, *mazF* was placed under the control of the trc riboswitch promoter and cloned into broad-host-range CRISPR/Cas12a vectors pSL2680 and pSEVA451. The transformed plasmids were then transferred to the wild-type marine *Synechococcus* strain RS9916 via conjugation. Development of these vectors will facilitate timely transformation of marine *Synechococcus* strains and will allow widespread use for studies related to photosynthesis, metabolism, synthetic biology, and biotechnology in cyanobacteria.

1. Knoot CJ, Ungerer J, Wangikar PP, Pakrasi HB. Cyanobacteria: promising biocatalysts for sustainable chemical production. *J Biol Chem.* 2018; 293(14):5044–5052. https://doi.org/10.1074/jbc.R117.815886
2. Ungerer J, Pakrasi HB. Cpf1 is a versatile tool for CRISPR genome editing across diverse species of cyanobacteria. *Sci Rep.* 2016; 6:39681. https://doi.org/10.1038/srep39681
3. Chen L, Liu H, Wang L, Tan X, Yang S. Synthetic counter-selection markers and their application in genetic modification of *Synechococcus elongatus* UTEX 2973. *Appl Microbiol Biotechnol.* 2021; 105. https://doi.org/10.1007/s00253-021-11391-y

Single-molecule studies of phycobilisome quenching with an OCP homolog

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Phycobilisomes (PBS) are dynamic antennae structures that facilitate light harvesting for photosynthesis in cyanobacteria by expanding the useable spectrum of light beyond the absorption of chlorophyll. Cyanobacteria inhabit diverse and often dynamically changing environments, so photoprotection is necessary during high light intensity that could damage this delicate photosynthetic apparatus. One of the main photoprotective mechanisms that cyanobacteria have evolved is a photoactivatable binding partner called the orange carotenoid protein (OCP). OCP binds to the core of the PBS and induces non-photochemical quenching, which prevents excess energy from reaching the reaction centers in the thylakoid membrane. Using an Anti-Brownian Electrokinetic (ABEL) trap, we have previously demonstrated that two photophysically distinct quenched states are detectable when quenching PBS with OCP, and that simulations can connect the experimentally observed photophysical states to the correct binding locations on the PBS. However, OCP is not the only quencher for PBS that is present in cyanobacteria. Here, we explore PBS quenching by OCP homologs to learn what other quenched states are created. We investigated how well the ABEL trap can differentiate between these quenched states to determine the limits of this approach for identifying binding sites and strengths of quenchers on the PBS.

Time-Resolved FTIR Difference Spectroscopy Reveals Altered Charge Recombination and Cofactor-Protein Interactions in Far-Red Acclimated Photosystem I

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Far-red light photoacclimation fundamentally remodels the pigment-protein architecture of Photosystem I (PSI), enabling the synthesis of chlorophyll f-containing complexes to harvest and utilize lower-energy photons beyond the conventional photosynthetic range. However, the effects of this remodeling on charge recombination dynamics and cofactor-protein interactions remain poorly understood. Using time-resolved step-scan FTIR spectroscopy at 77 K, we compared PSI complexes isolated from Fischerella thermalis PCC 7521 cells grown under white-light (WL) and far-red-light (FRL).

Global analysis of the time resolved spectra uncovered a rapid contribution with signatures of a laser-induced heating artefact, and a slower component assigned to $P700^{+}A_{1}^{-}$ charge recombination. Charge recombination was markedly slower in FRL-PSI, exhibiting a lifetime of ~626 μs compared with ~345 μs in WL-PSI. The latter time constant is similar to that found in PSI samples from Synechocystis sp 6803. Factors that could give rise to this more than 1.8-fold increase in charge separation lifetime in FRL-PSI will be discussed.

Close association of PsbO, a thylakoid lumen protein, and transcription factor protein Sll0176, as revealed by structural mass spectrometry

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PsbO (often termed the **Manganese-Stabilizing Protein**, or **MSP**, ~33 kDa) is the core extrinsic luminal subunit of Photosystem II (PSII) [1]. Encoded by the single-copy gene *PsbO*, e.g., *sll0427* in *Synechocystis* sp. PCC 6803, it is the only extrinsic subunit conserved across all oxygenic phototrophs, from cyanobacteria to glaucophytes, red algae, and green plants, or so-called oxygenic photosynthetic organisms. PSII splits water (H_2O) to release electrons for linear electron transport and the protons released for the proton gradient establishment that drives ATP synthesis; the oxygen is released as a byproduct, O_2, which has revolutionized life history. Using the membrane-permeable cross-linker DSS (disuccinimidyl suberate), we have performed *in vivo* cross-linking of proteins in structural proximity. We found that PsbO, a luminal protein, is cross-linked to a transcription factor protein, Sll0176 (PsbO_K^{89}-Sll0176_K^{104}), as supported by high-confidence MS1 and MS2 spectra. Sll0176 is a basic protein and is annotated as a nickel-responsive DNA-binding transcription repressor that negatively regulates DNA-templated transcription. The crystal structure of the Sll0176 protein has been partially resolved in previous research (PDBID:5FMN) [2]. We have used our structural mass spectrometry data to guide the structural interactions of PsbO and Sll0176. Our results suggest that cyanobacterial PsbO may play dual roles in regulating gene expression under heavy metal stress and PSII activity.

[1] Y. Umena, K. Kawakami, J.-R. Shen, N. Kamiya, Crystal Structure of Oxygen-Evolving Photosystem II at a Resolution of 1.9 Å, ***Nature***, 473 (2011) 55-60

[2] A.W. Foster, R. Pernil, C.J. Patterson, A.J.P. Scott, L.-O. Pålsson, R. Pal, I. Cummins, P.T. Chivers, E. Pohl, N.J. Robinson, A Tight Tunable Range for Ni(Ii) Sensing and Buffering in Cells, Nature Chemical Biology, 13 (2017) 409-414

Inefficient Carotenoid-Mediated Quenching of Chlorophyll *a* Triplet in IsiA: A Hypothetical Cause for the Evolution of Cysteine-Mediated Quenching of Chl *a* Excited Singlet.

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Iron-stress-induced protein A (IsiA) is a chlorophyll (Chl) *a*-containing protein in cyanobacteria, produced and accumulated in large quantities during naturally prevailing low-iron growth. IsiA shares high homology with the intrinsic antenna of Photosystem II, CP43, and likely both complexes co-evolved from a common ancestral protein. Because of their structural similarity, IsiA and CP43 share similar basic spectroscopic properties. When expressed, IsiA forms IsiA-only and photosystem I (PSI)-IsiA complexes. The IsiA-only complex shows a cysteine-mediated photoprotective mechanism that significantly quenches Chl *a* singlet excitation, detectable as a decrease in fluorescence lifetime, and partially mitigates the risk of photo-oxidative damage of quenched Chl *a*. However, the effectiveness of longer-time-scale photoprotection, such as Chl *a* triplet quenching via Chl *a* $\rightarrow$ carotenoid triplet-triplet sensitization (TTS), remains unknown. Our evaluation suggests an extraordinarily slow TTS with a time constant of 100 ns, which is comparable to the concurrent sensitization of singlet oxygen. From this perspective, the photoprotective role of carotenoids in IsiA is greatly underperforming. Therefore, a cysteine-based Chl *a* singlet quenching could be seen as a supplementary photoprotective mechanism, developed specifically to complement the underperforming Chl *a* $\rightarrow$ carotenoid TTS.

This study is supported by funding from the U.S. Department of Energy (DOE), Office of Basic Energy Sciences, through the grant DE-FG02-99ER20350 to HBP.

Unraveling Decoherence Mechanisms of Electron Spin Qubit Pairs in Photosynthetic Reaction Centers

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Photosynthetic reaction center (RC) proteins serve as well-defined, experimentally tunable platforms for investigating quantum spin dynamics. Light-induced sequential electron transfer within these RCs generates long-lived, spin-correlated radical pairs (SCRPs)—also recognized as spin qubit pairs (SQPs)—which have provided critical insights into photosynthetic structure-function relationships and emerge as promising candidates for quantum sensing. However, the precise factors governing the loss of coherence in the protein matrix remain unresolved.

In this work, we combine experimental and computational methods to elucidate coherence times and decoherence pathways in Photosystem I (PSI) and type II RCs of purple bacteria (bRCs). High-frequency/high-field electron paramagnetic resonance (EPR) spectroscopy operating at D-band (130 GHz , 4.6 T) was employed to measure coherence decay times (T_M) via two-pulse electron spin echo and Rabi oscillation experiments across a wide range of temperatures. Contrary to conventional models where proton-mediated nuclear spin diffusion dominates low-temperature decoherence, deuteration yielded only a negligible to modest increase in coherence times across both PSI and bRCs. Furthermore, coherence properties displayed minimal sensitivity to biochemical modifications and biological species.

The experiments combined with spin-dynamics simulations demonstrate that classical nuclear spin diffusion and instantaneous/spectral diffusion mechanisms alone cannot account for the measured coherence decay times. Instead, we identify low-temperature quantum tunneling of methyl groups in the microenvironment of the unpaired spin centers as the primary drivers of decoherence. These findings provide new insight into electron spin decoherence processes in proteins and offer key principles for engineering quantum sensing systems.

Explicit Inclusion of Vibronic Effects in Modeling the Optical Absorption Spectra of FMO

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Photosynthetic pigment-protein complexes transfer excitation energy with efficiencies that bio-inspired materials still need to match. The efficiency of excitation-energy transport in photosynthetic complexes is argued to critically depend on electron-vibrational (vibronic) interactions. However, quantifying the role of vibronic interactions is challenging because pigment-protein complexes often have congested optical spectra that include contributions from individually coupled pigments and the proteins they are embedded in, making it difficult to disentangle effects and provide needed insight. From a computational perspective, the Hamiltonian used to model these processes, accounting for relevant electronic and vibrational degrees of freedom of photosynthetic pigments, grows exponentially with the size of the vibrational basis. Various perturbative and numerical methods have been proposed, but the challenge remains. In this work, we introduce an efficient approach to construct a compact vibronic Hamiltonian based on the Foulton-Gouterman electron-vibrational model. Using an importance estimator based on the contribution of individual vibrational modes to the overall vibronic spectra, we significantly reduce the size of the Hilbert Space and, in effect, the size of the vibronic Hamiltonian. Using this idea, we model the absorption and circular dichroism (CD) spectra of the Fenna-Matthews-Olson (FMO) complex, showing excellent agreement with experiment and directly demonstrating how vibrational degrees of freedom contribute to the optical spectra in this prototypical photosynthetic model system.

Temperature Dependence of Absorption and Fluorescence Spectra in Water-Soluble Chlorophyll Protein Using Variational Polaron Transformation

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Excitation energy transfer in pigment-protein complexes is governed by a delicate interplay between electronic coupling between chromophores, vibrations in the protein lattice, and temperature-dependent system-bath interactions. Previous work by Renger and Marcus used a second-order cumulant expansion with the partial ordering prescription (POP) to describe optical lineshapes, including vibrational sidebands and broadening caused by exciton relaxation[1]. Complementary work by Jang developed polaron-transformed master equations to describe excitation energy transfer across weak and strong coupling to the vibrational environment[2,3]. In this work, we apply a second-order POP cumulant expansion in the variational polaron frame to *Lepidium virginicum* water-soluble chlorophyll protein (Lv-WSCP) and evaluate the bath correlation functions arising from residual system-bath interactions. We calculate population-transfer rates between exciton states and incorporate their contributions into absorption and fluorescence lineshape calculations. Our calculations indicate increasingly Förster-like transfer dynamics at higher temperatures and more Redfield-like dynamics at lower temperatures. These results connect temperature-dependent exciton relaxation to the absorption and fluorescence spectra of Lv-WSCP.

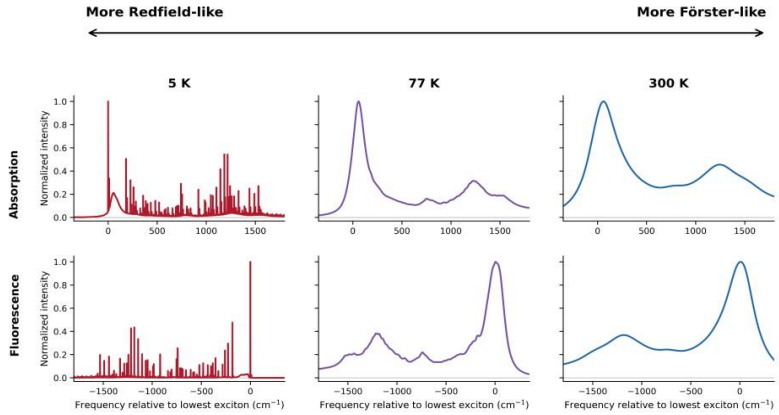


Figure 1. Absorption and fluorescence lineshapes of wild-type Lv-WSCP in the polaron frame, including contributions from the cumulant expansion of off-diagonal population transfer. Spectra are shown at 5, 77, and 300 K. At lower temperatures, the lineshapes exhibit more Redfield-like characteristics, whereas increasing temperature drives the system toward a more Förster-like regime.

1 T. Renger and R. A. Marcus, "On the relation of protein dynamics and exciton relaxation in pigment–protein complexes: An estimation of the spectral density and a theory for the calculation of optical spectra," J. Chem. Phys. 116, 9997–10019 (2002). doi:10.1063/1.1470200.
2 S. Jang, Y.-C. Cheng, D. R. Reichman, and J. D. Eaves, "Theory of coherent resonance energy transfer," J. Chem. Phys. 129, 101104 (2008). doi:10.1063/1.2977974.
3 S. Jang, "Theory of multichromophoric coherent resonance energy transfer: A polaronic quantum master equation approach," J. Chem. Phys. 135, 034105 (2011). doi:10.1063/1.3608914.

Sr(-ious) Substitution: How Strontium Alters Electron Transport in Photosystem II of *Chlorella ohadii*

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The role of calcium in the water-oxidizing complex (WOC) of Photosystem II (PSII) is established in supporting the structure and function of the manganese oxidation cycle, with strontium (Sr^{2+}) being the only known functional substitute. *Chlorella ohadii* was used as a model green alga because of its high photosynthetic efficiency and light tolerance, making it suited for investigating the physiological effects of metal substitution *in vivo*. Biosynthetic substitution of Ca^{2+} with Sr^{2+} was investigated to determine how Sr^{2+} alters photosynthetic electron transport. 77K fluorescence spectroscopy revealed an increased PSII:PSI ratio in Sr^{2+}-substituted cells, along with increased carotenoid-associated energy dissipation, indicating reduced photochemical use of excitation energy and greater diversion of excess energy toward photoprotection. Fast repetition rate fluorometry demonstrated diminished period-four oscillations and an increased beta parameter, consistent with less efficient WOC cycling and peroxide-associated decay of the complex. Q_A^- reoxidation kinetics indicated relatively unaffected Q_A to Q_B electron transfer rates but an increased proportion of inactive PSII centers and predominantly oxidized Q_B, suggesting disrupted downstream electron utilization. Cytochrome b_6f redox kinetics showed greater oxidation of cytochromes b and f under Sr^{2+} substitution, while plastocyanin was oxidized less by PSI. P700 measurements also demonstrated reduced overall PSI activity, consistent with limited electron supply from PSII, while electrochromic shift measurements indicated weaker proton-gradient formation. Together, these results indicate that Sr^{2+} substitution destabilizes WOC turnover in *C. ohadii*, impairing PSII function and creating electron transport limitations across the photosynthetic electron transport chain, reducing overall photosynthetic efficiency.

The Regulation of C4 photosynthesis in Field Grown Sorghum Plants

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Field environmental conditions experienced by crops vary dramatically in response to daily cycles as well as sporadic events such as cloud cover or extreme weather events. To date, most research on plant responses to abiotic stress has been conducted under controlled indoor conditions. To extend laboratory research towards improvement of agricultural yield, it is essential to understand how plants respond to real-world field conditions. C4 plants are generally regarded as more stress tolerant than C3 plants due to the separation of the photosynthetic electron transport chain (PETC) and carbon fixation into the mesophyll and bundle sheath cells respectively. This altered anatomy allows for carbon to be concentrated in the bundle sheath cells, limiting photosynthetic loss to photorespiration. The impact that this spatial separation has on source/sink balance of the PETC and carbon fixation in the turbulent field setting is largely unstudied, despite C4 crops such as *Zea mays* (maize), *Sorghum bicolor* (sorghum), and *Saccharum officinarum* (sugarcane) being major feedstocks. Here, we address this knowledge gap by evaluating how the PETC activity of field grown sorghum is impacted by leaf developmental stage, canopy level, and time of a day. To determine the impact of light dynamics throughout the day and canopy shading on PETC, a MultispeQ 2.0 was used to measure photosynthetic activity at 9am, 1pm, and 5pm and across three leaf classes top young (TY), top mature (TM), and bottom mature (BM) in field-grown sorghum leaves (Fig. 1). These three leaf classes were chosen based on differences in received light and carbon assimilation capacities. We hypothesize that light intensities and developmental stages/canopy level impact C4 photosynthesis. We found that PSII activity is highest in the morning when temperature and light levels are the lowest. Additionally, PSI cyclic electron flux (CEF), an alternate electron flux pathway that provides photoprotection and produces ATP, increases at the midday time point. BM leaves, despite having high chlorophyll contents and photosynthetic capacity, have the lowest photosynthetic activity of the three developmental stages. Further work will be done to evaluate gene expression, carbon metabolism, and structural alterations in sorghum plants across these timepoints and leaf classes. Together this work will be used to identify bottlenecks in sorghum which can be identified as potential engineering targets to improve crop yields under real-world field conditions.

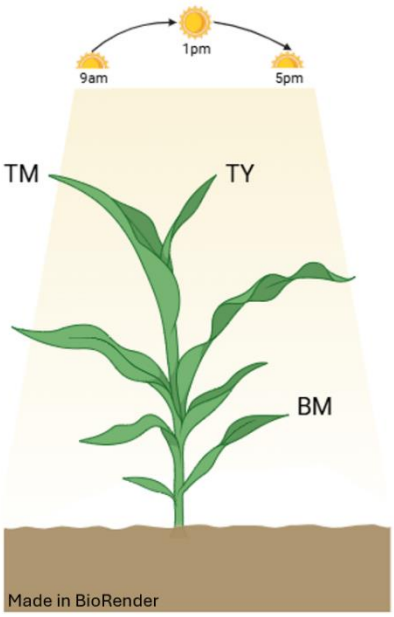


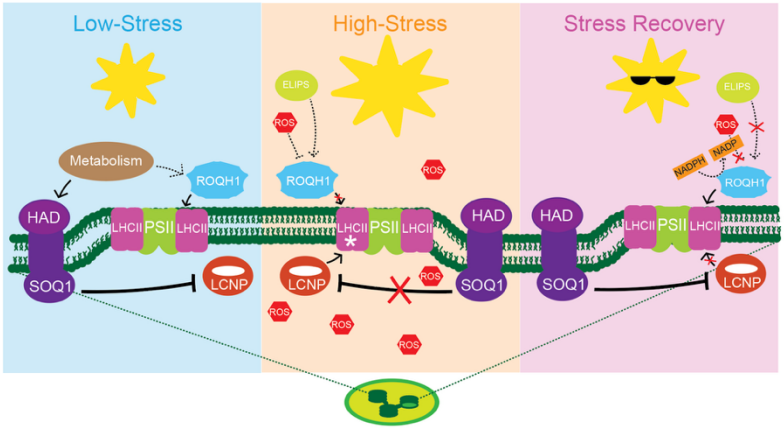
Figure. 1. MultispeQ measurements were taken across time points (9am, 1pm, and 5pm) and leaf developmental stages (top young-TY, top mature-TM, and bottom mature-BM) to determine the impact of light levels and canopy shading on photosynthetic electron transport chain activity.

Illuminating the Function of Early Light Induced Proteins in qH Non-Photochemical Quenching

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The fundamental mechanism of photosynthesis involves a delicate balance between energy usage and energy dissipation. Enough energy must be absorbed for photochemistry, but excess energy can result in the generation of reactive oxygen species (ROS), which may damage the photosynthetic apparatus. Therefore, excess energy is dissipated as heat through the process of non-photochemical quenching (NPQ). An existing gap in our understanding of photoprotection is the mechanism and control of sustained quenching (qH), a slow-acting component of NPQ. To understand qH, we must identify its effectors and define the key molecular players involved in its activity, regulation, and integration into larger metabolic pathways. We have established that qH is slowly alleviated by the stromal protein RELAXATION OF QUENCHING 1 (ROQH1) and prevented by the transmembrane protein SUPRESSOR OF QUENCHING 1 (SOQ1). It is induced by the lumenal protein LIPOCALIN IN THE PLASTID (LCNP).



Schematic depicting known mechanisms of qH NPQ (solid arrows) and potential models for ELIP involvement (dashed arrows.

Though NPQ is known to occur in response to light-generated signals, such as protonation for qE, it remains unclear which signals are sensed and integrated for the repression and activation of qH by ROQH1 and LCNP, respectively. EARLY LIGHT INDUCED PROTEINS 1 and 2 (ELIPS 1 and 2) are intriguing potential interaction partners of ROQH1. ELIPS 1 and 2 are membrane proteins with three transmembrane alpha-helices and four putative chlorophyll binding sites. They have been shown to accumulate diurnally, during seedling etiolation, and in response to light stress. However, their accumulation has not been universally or directly connected to a photoprotective function.

ELIPS were associated with LHCII when they were first discovered in *Pisum sativum*, and later research has found that ELIPS associate with LHCII in *Arabidopsis thaliana* under stress conditions. Furthermore, the Chlamydomonas homolog of ELIPs 1 and 2, Msf1, has been reported to play a role in stabilizing protein-chlorophyll interactions, including those in light harvesting complexes, particularly in response to stress. Our lab has demonstrated that qH occurs in the antennae of PSII, and that it may involve a chlorophyll-chlorophyll excitonic interaction.

Here we offer preliminary results indicating that ELIP1 and ELIP2 expression is upregulated in *soq1 roqh1* mutants and propose models regarding how this interaction could help activating or inactivating qH.

Spectroscopy of reactive intermediates and photoanode activity of Mn4V4 water oxidation catalyst mimicking oxygen evolving complex of Photosystem II

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Light driven water oxidation is taking place in the Mn_4Ca oxygen evolving complex (OEC) of Photosystem II. Molecular manganese vanadium oxide cluster $[Mn_4V_4O_{17}(OAc)_3]^{3-}$ was previously reported as a synthetic model of the OEC which catalyzes the homogeneous water oxidation under selected conditions. Initial complex has $Mn_2^{III}Mn_2^{IV}$ electronic structure analogous to S_1-state of the Photosystem II OEC. Here we obtained *in situ* Mn K-edge XAS spectroscopy of the electrochemically oxidized intermediates analogous to S_2 and S_3 states as well as activated reaction intermediate featuring all Mn^{IV} electronic structure and short Mn^{IV}=O (~1.65 Å). Due to its extremely low solubility in water, a thin, highly stable film of the $[Mn_4V_4O_{17}(OAc)_3]^{3-}$ can be formed on the FTO/or ITO electrodes where it works as a photoanode producing ~$1*10^{-4}$ A/cm^2 photocurrent with applied bias of ~1.4 V vs Ag/AgCl in 0.1M HNO_3. Data are compared to other reported photo-anodes active in acid. Similarities and differences with Photosystem II will be discussed.

Towards the Examination of Early Subunit Binding of the Alpha-Carboxysome Shell

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The alpha-carboxysome is a large (> 100 MDa, 100 nm) proteinaceous microcompartment composed of 1000s of individual subunits that encapsulates RuBisCO to isolate the CO_2 fixation reaction from O_2. Prior work has identified the key subunit interactions responsible for shell formation, but the role of several well-conserved but non-essential motifs on the disordered scaffold CsoS2 within the alpha-carboxysome has not yet been understood. Assembly intermediates between CsoS2 and shell subunits have never been isolated in vitro, obscuring how these motifs might regulate alpha-carboxysome assembly. We present an anti-Brownian electrokinetic (ABEL) trap for long-term observation of small, heterogeneous collections of protein subunits in vitro at biologically relevant protein concentrations. Binding between shell subunits and truncated scaffold protein constructs will be assessed by FCS to distinguish binding from nanoscale aggregation. ABEL trapping will extract transient binding pathways and stoichiometries.

Comparative Effects of Anion Supplementation on the Acceptor Side of Photosystem II

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In Photosystem II (PSII), the non-heme iron (NHI) stabilizes Q_A and Q_B redox states, which mediates the electron transfer between plastoquinone acceptors, and regulates recombination pathways. Because the NHI is ligated by bicarbonate and amino acid residues, its coordination environment is strongly influenced by ionic composition, making it a key regulatory site for photosynthetic electron transport. *Picochlorum oklahomense,* a halotolerant green alga capable of thriving across a wide salinity range, provides a model system to examine how anions influence PSII function. To study ion-specific effects, cultures were supplemented with anions spanning the spectrochemical series (HCO_3^-, Cl^-, Br^-, I^-, F^-, NO_2^-, SO_3^{2-}) and their impact on Q_A^- redox kinetics and charge recombination were analyzed using chlorophyll fluorometry. Q_A^- redox analysis showed that addition of halides caused a decrease in the Q_A^--Q_B^- transfer time, suggesting a facilitation of electron transfer due to modulation of either protonation of the semiquinone or local redox poise. Nitrite and sulfite slowed electron transfer, with sulfite additionally limiting the Q_A^--Q_B transition. DCMU inhibitor studies suggest that halide addition restored Q_A^- reoxidation significantly the harder of a Lewis base was added. Negative impact was observed with NO_2^- and SO_3^{2-} suggesting localized environmental shift within the protein matrix that increased the accessibility or affinity of the Q_B site for DCMU. Fast repetition rate fluorometry studies suggest that strong field-directing ions caused limitation of quinone access to the Q_B site. The mildly beneficial halide effect on electron transfer may be localized to the depletable bicarbonate side responsible for protonation of Q_B. Nitrite appears to limit Q_B turnover, while sulfite fully inhibits the operation of PSII. The totality of sites of activity of sulfite are not clear, but Q_A^- redox modulation supports that it is affecting the acceptor side. DCMU inhibitor studies appeared to preserve fluorescence quality and counteract DCMU interference among halides. NO_2^- and SO_3^{2-} data suggest more complex interactions involving NHI and Q_A^- reoxidation. High salt concentrations were seen to block DCMU binding effectively, indicating importance of ionic environment for stabilizing PSII electron transfer. Rate oximetry suggests that halide addition increases the rate of oxygen production overall. However, SO_3^{2-} resulted in full loss of cell viability. NO_2^- increased net oxygen production, yet also increased oxygen consumption.

A Glycolytic Enzyme Moonlighting in Photosynthesis

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Entner-Doudoroff (ED) aldolase (Eda), a key enzyme traditionally associated with the glycolytic ED pathway, is highly conserved among oxygenic phototrophs. Recent evidence shows that the ED pathway is absent in cyanobacteria and plant chloroplasts, with Eda likely serving as an oxaloacetate (OAA) decarboxylase that produces pyruvate from photosynthate-derived OAA. However, the role of Eda in oxygenic phototrophs remains elusive. In *Synechococcus elongatus* PCC 7942, the Δ*eda* mutant shows severe chlorosis after extended growth under standard light, suggesting a role in photosynthetic light energy harvesting. Here, we show that Eda is involved in photosystem I (PSI) assembly and function. The Δ*eda* mutant shows reduced photosystem I (PSI) monomer abundance but increased ATP synthase levels across various light conditions. Consistent with this, Δ*eda* shows higher $P700^+$ re-reduction rates than the wild type, suggesting higher cyclic electron flow. TurboID-based proximity labeling further places Eda near important PSI assembly factors. Lastly, photosystem proteins were more abundant in the soluble proteomic fraction of the mutant, suggesting that photosystem components remain largely unassembled in the absence of Eda. Together, our results reveal a new role for a conserved carbon metabolic enzyme in photosynthesis, providing new insights into photosynthesis regulation.

Metal-Dependent Tuning of the Plastocyanin Entatic Coordination Core Probed by d_2Cys and d_3Met Vibrational Reporters

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Metal substitution in plastocyanin perturbs the Cu site in ways that are reported differently by local CD vibrational probes at Cys89 and Met97. Using new d_2Cys89 data for Co(II)-substituted Nostoc Pc and Pc-Cyt *f* complexes, analyzed alongside previously reported Cu(I)/Cu(II)/Zn(II) d_2Cys and d_3Met series produces a general increase in the symmetric d_2Cys and d_3Met frequencies, but that Co(II) introduces an additional ~17 cm^{-1} separation in d_2Cys relative to Cu(II) and Zn(II) that is not mirrored by d_3Met. We further characterize Pc-Cyt *f* complexes containing Co(II)Pc and Cu(II)Pc, finding that complexation of Co(II)Pc produces only a small d_2Cys89 red shift, comparable to the ZnPc response, whereas native CuPc complexes show a large d_2Cys89 blue shift. Complementary UV-vis measurements on Cu(II)Pc-Cyt *f*(III) reveal a pronounced red shift and intensity decrease of the Pc ligand-to-metal charge-transfer band upon complexation, underscoring that the vibrational and electronic probes report consistent but distinct aspects of Pc-Cyt *f* entatic-state remodeling. Density functional theory calculations on a first-shell Pc model show that frozen-geometry metal substitution at the Cu(I) and Cu(II) structures yields only modest Co-specific differences, indicating that electronic identity at a common geometry cannot account for the Co anomaly. Systematic scans of metal-ligand bond lengths, donor-metal-donor angles, dihedrals, and scaffold-like Cys displacements reveal that d_2Cys is tuned by collective remodeling of the Cu-Cys-Met-His coordination core, characterized by a crossed-edge hinge between the Cys/His39 and Met/His92 sectors, and by local gating through the Cys rotamer and $C_\beta D_2$ geometry. These structural pathways rationalize the Co(II)-specific d_2Cys blue shift and place metal substitution and Pc-Cyt *f* complexation within a common framework for entatic-state tuning in blue copper proteins.

Partitioning of Abiotic Reaction Pathways with Bacterial Microcompartment Protein Shells

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Bacterial microcompartments (BMCs) are selectively permeable protein shells that encapsulate enzymes and substrates to increase catalytic throughput of metabolic pathways via means of confinement. Compartmentalization of reaction pathways is a ubiquitous phenomenon in biological systems, but more difficult to achieve through synthetic means. As such, in this study the self-assembling BMC architecture is exploited for compartmentalization of an abiotic reaction to leverage the chemistries of synthetic molecules within biological microenvironments. A benchmark multimolecular system for photocatalytic hydrogen evolution was employed comprised of Ru(2,2'-bipyridine)$_3$ photosensitizer, chloro(pyridine)cobaloxime catalyst, and ascorbate sacrificial donor. The biohybrid shells successfully supported hydrogen evolution and, under certain conditions, enhanced catalytic turnover relative to the bulk reaction without protein. This combined experimental and computational study provides important insights into the mechanisms by which BMCs enhance catalysis, focusing on selective permeability both as a means to colocalize as well as partition reactive intermediates. These results highlight the potential for programming BMC shells with synthetic capabilities for energy applications.

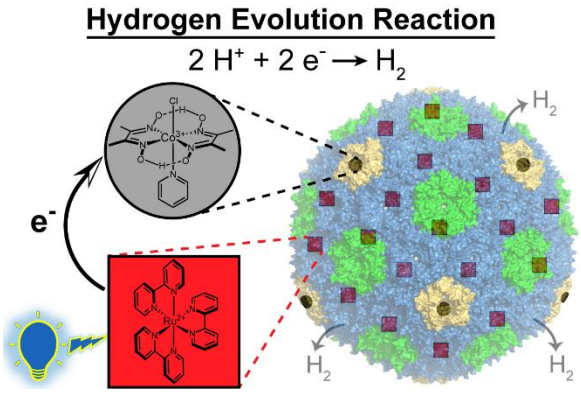


Figure. 1. BMC protein shells loaded with synthetic catalysts and photosensitizers for light-driven hydrogen evolution.

Impact of Constitutive Overexpression of Glutathione S-Transferase (GSTS1) on Photosynthesis in *Chlamydomonas reinhardtii*

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Photosynthetic organisms like the green alga *Chlamydomonas reinhardtii* frequently face stressors such as high light and low temperature, which overexcite the thylakoid membranes and generate reactive oxygen species (ROS), including destructive singlet oxygen ($^1O_2$). Under normal conditions, Photosystem II (PSII) splits water and passes electrons through the cytochrome b6f complex to build a proton gradient that powers the cell. This study examines a genetic mutant of the alga, (CC-4610), which constitutively overexpresses GSTS1 under the PSAD promoter, comparing it to wild-type (CC-125) through a spectroscopic analysis. We determined how GSTS1 overexpression affects ROS mitigation, electron transport kinetics, ATP synthesis, and overall energy conversion efficiency in *C. reinhardtii*. Wild-type and mutant strains were compared for efficiency under high light across photosynthetic electron transport processes and growth. Compared to wild-type, CC-4610 is a rare mutant that grows faster. Q_A^- reoxidation measurements showed higher reduction and increased turnover of the plastoquinone pool in the mutant, and ECS measurements showed a much larger proton motive force than in wild-type. Together, these results point to electrons being diverted from the electron transport chain to the mitochondria in CC-4610, boosting respiration and allowing further water oxidation to take place, preventing photodamage at PSII. That, combined with higher ATP production, likely explains the mutant's growth increase.

Real-Time TOC Receptor Mobility Reveals Dynamic Import Microdomains in Isolated *Pisum sativum* Chloroplasts

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Chloroplast biogenesis and photosynthetic performance require targeted import of ~2,700 nuclear-encoded proteins from the cytosol. This trafficking is mediated by the TOC/TIC translocon system, which must deliver high throughput while maintaining selectivity across the chloroplast double membrane. Although core TOC components (e.g., Toc75, Toc34, Toc159) and key regulatory inputs (ATP/GTP, transit peptides, receptor interactions, and proteolytic remodeling) have been identified, how the import machinery is organized in space and time in intact plastids remains unclear. Most mechanistic models derive from *in vitro* assays or partially solubilized components, leaving a gap in understanding whether TOC/TIC behaves as uniformly distributed channels or as a spatially compartmentalized, dynamically assembling system coordinated at envelope microdomains and contact sites.

Here we interrogate TOC organization and dynamics in intact, isolated Pisum sativum chloroplasts, focusing on the receptor GTPase Toc34 as an entry point for mapping import-site behavior. Using Laser scan and Airy scan immunofluorescence confocal microscopy and immunogold electron microscopy, we observe Toc34 concentrated in discrete punctate envelope sites, consistent with specialized translocon microdomains and potential contact regions that could organize TOC/TIC coupling. We further examine Toc34 co-localization with Toc159 and Toc75 across plastid maturation, revealing patterns consistent with developmentally regulated remodeling of import-site composition or abundance. To connect localization to function, we apply single-particle tracking by TIRF microscopy, resolving two Toc34 mobility states: a faster, freely diffusing population and a slower, confined-yet-dynamic population. This switching is consistent with Toc34 cycling between an unengaged diffusion mode and a transiently stabilized state associated with receptor clustering, precursor engagement, and/or coupling to translocation.

Future directions will build a mechanistic bridge from these in situ observations to import function. We will test whether Toc34 confinement correlates with active import by adding defined precursors/transit peptides and manipulating nucleotide state (ATP/GTP/GDP analogs) while quantifying mobility shifts and microdomain stability. We will expand multiplex imaging to additional TOC/TIC markers and candidate contact-site factors to determine whether puncta represent TOC–TIC hubs or modular receptor assemblies, and will examine how import-site number, composition, and dynamics scale across controlled plastid maturation stages. Finally, we will pursue correlative super-resolution/EM approaches to link single-molecule dynamics to ultrastructural context at defined envelope regions.

Toward the structural basis of far-red light photoacclimation in Photosystem I

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Most oxygenic photosynthetic organisms primarily use chlorophyll *a* to absorb visible light. Some cyanobacteria, however, can acclimate to environments enriched in far-red light by remodeling their photosynthetic machinery and synthesizing the red-shifted pigments chlorophylls *d* and *f*. This process, known as far-red light photoacclimation (FaRLiP), expands the wavelengths of light that can be used for photosynthesis and provides a model for understanding how photosynthetic organisms adapt to different light environments. Our research investigates FaRLiP in cyanobacteria using complementary physiological, spectroscopic, and structural approaches. Controlled-light growth experiments are being used to examine how cyanobacteria respond to white and far-red light, including changes in growth, pigmentation, and light absorption during acclimation. In parallel, structural studies are being used to examine how far-red-light-absorbing chlorophylls are incorporated into photosystems and how their local protein environments may contribute to their function. These approaches connect organism-level responses to light with molecular changes in the photosynthetic machinery. Understanding the mechanisms that allow cyanobacteria to use wavelengths beyond the range employed by most oxygenic phototrophs provides insight into the adaptability of photosynthesis and may ultimately inform efforts to engineer photosynthetic systems with altered light-harvesting properties.

Metabolic crosstalk: Exploring sugar-mediated feedback control on plant photoprotection

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Abstract

Photosynthesis harnesses light energy in chloroplasts, which supports plant productivity, food security, and climate regulation. Yet excessive light can impair it by generating reactive oxygen species (ROS) and increasing photorespiration by favoring ribulose-1,5-bisphosphate (RuBP) oxygenation over carboxylation by ribulose-1,5-bisphosphate carboxylase/oxygenase (Rubisco). Although this reaction mitigates chloroplastic ROS, it produces the toxic metabolite, 2-phosphoglycolate (2PG), which is detoxified by 2-phosphoglycolate phosphatase (PGLP) through dephosphorylation. Plants dissipate excess light as heat through non-photochemical quenching (NPQ), including qH, a slowly reversible, LIPOCALIN IN THE PLASTID (LCNP)-mediated photoprotective mechanism. SUPPRESSOR OF QUENCHING 1 (SOQ1) negatively regulates qH via its lumenal domains by inactivating LCNP, whereas RELAXATION OF QH1 promotes qH relaxation. The function of the stromal haloacid dehalogenase-like (HAD) domain of SOQ1 remains unknown. Interestingly, *SOQ1* also produces an isoform that encodes a soluble HAD-like domain (SOQ1 HAD) in the stroma. Preliminary findings suggest SOQ1 HAD co-expresses with PGLPs, contains a sugar monophosphate-binding pocket, and exhibits phosphatase activity toward the monophosphate ribose-5-phosphate (R5P), which is involved in RuBP regeneration. This phosphatase activity is reduced in the catalytically inactive SOQ1 HAD^{D80N} variant. Intriguingly, R5P can competitively inhibit PGLP, suggesting another potential connection between carbon metabolism and photorespiratory flux. We hypothesized that, under stress conditions that favor the Rubisco oxygenation reaction, SOQ1 HAD dephosphorylates accumulating sugar phosphates that would otherwise inhibit PGLP and suppress photorespiration. This would allow photorespiration while triggering SOQ1 HAD to prevent the interaction between SOQ1 lumenal domains and LCNP, thereby inducing protective NPQ, consistent with our observation of LCNP-dependent NPQ increase during photorespiration in *Arabidopsis thaliana*. qRT-PCR shows that the transcript isoform encoding for the soluble SOQ1 HAD domain remains stable under high-light and non-stress conditions in wild-type *A. thaliana*. We are currently determining protein levels, whether soluble HAD accumulates, and its location in the chloroplast. To define SOQ1 HAD substrate specificity and activity, we are generating lines expressing SOQ1 lacking the HAD domain, HAD alone, and the catalytically inactive SOQ1 HAD domain variant. We propose that SOQ1 HAD may have a dual role in metabolic regulation and photoprotection, promoting crosstalk between the stroma and lumen; investigating this crosstalk will provide insights into the SOQ1 HAD function and how plants adapt photosynthesis to stress, informing strategies to develop climate-resilient crops.

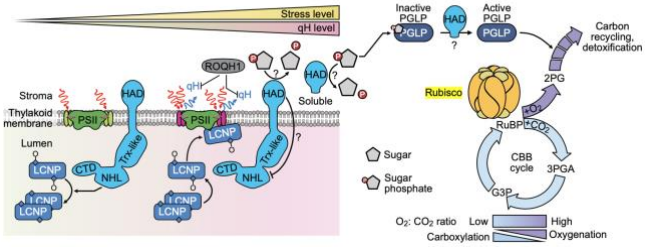


Fig.1. SOQ1 HAD-like domain sugar-phosphatase activity may link photoprotection and photorespiration.

Hands-free Photosynthesis: Autonomous Leaf-Level Multi-Chamber Gas Exchange System

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We introduce a novel automated multi-chambered gas exchange system for non-destructive autonomous measurements of leaf physiology, addressing a current phenotyping bottleneck. Omics approaches have advanced significantly in the last decade, but progress in the phenotyping of physiological traits has lagged behind, with negative consequences for the pace of crop improvement. Here we describe and evaluate an Automated Photosynthesis System (APS) that enables autonomous, repeated, *in situ* measurements of photosynthetic CO2 assimilation (*A*), stomatal conductance (*gs*) and transpiration (E) across multiple leaves near simultaneously. The APS integrates a single infrared gas analyser with multiple independently controlled chambers that operate sequentially, allowing instantaneous or 'snapshot' measurements of leaf photosynthetic parameters to be captured over space and time. System performance was assessed under controlled and greenhouse environments using two plant species (cowpea and sugarcane) while using commercially available systems to provide baseline performance for comparison. The APS produced consistent and repeatable measurements between chambers and over time, with measurements strongly correlated with a commercial Infra-red gas analysis system. Measurements of diurnal responses to changes in light intensity aligning closely with measurements from a and whole-plant transpiration system, confirming the accuracy of the system. Spatial variation in gas exchange within the canopy was also resolved, with higher rates observed in upper canopy leaves compared to lower leaves. Overall, the APS system described here, enables increased temporal and spatial resolution of gas exchange measurements, while also reducing labour and equipment constraints. This system provides a practical and cost-effective approach for integrating direct physiological measurements into high-throughput phenotyping pipelines, facilitating improved understanding of plant performance under realistic conditions.

Title: Mechanisms of Photosystem II Inhibition by Antimycin A in *Chlorella ohadii*

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Photosystem-II CEF (PSII cyclic electron flow) is an essential component of the growth of extremely fast-growing microbial phototrophs, including crucial biofuel producers. Antimycin A (AA) has been known as an inhibitor of electron transport in photosystems. By conducting inhibitor studies of PSII cyclic electron flow using Antimycin A in the PSII-CEF model organism *Chlorella ohadii*, mechanisms and key inhibitory binding sites can be identified. Monitoring the redox state of the redox-active components of cytochrome b_6f shows Q cycle and CEF redox signals to track the movement of electrons in the electron transport chain (ETC). The low and high light adapted samples with AA kept in the dark before fluorometry measurements did not indicate any changes to the apparent ratio of PSII to PSI. The light exposure after introduction of inhibitor allowed for rebalancing of the ETC, especially impacting PSI oxidation, due to competition of AA for quinone binding sites. AA preferentially inactivates inefficient reaction centers, resulting in better S-state oscillations and fluorescence yield under high light growth conditions. The ratio of PQ:PSII is diminished under low light treatment with AA but no such effect is seen under high light incubation. Altering the amount of available plastoquinone per PSII can alter the rate of facilitated movement of electrons within the complex. PyMOL (PDB: 4V62) measurements and AlphaFold3 docking show key amino acids interacting with plastoquinone (PL9) head group and AA bound in Q_B. These results suggest that the photoprotective mechanisms regulating the Q_B site in extreme light conditions can also protect PSII from AA-induced degradation.

Cyanobacterial Metabolite Overflow and Potential Forensic Application: Identification and Time Interval Estimation

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Cyanobacteria are aquatic microorganisms that can adapt their metabolism to environmental changes and participate in important biogeochemical cycles on a global scale, such as carbon and nitrogen cycles. An important response to environmental stress is the accumulation of cellular energy and carbon in excretory metabolites, metabolites produced and released extracellularly (Sebesta et al 2024). Knowing this process can help gain insight into the management of cyanobacterial energy and yield novel applications in forensic science. The goal of this work is to characterize metabolite leaks in cyanobacterium Synechocystis sp. PCC 6803. Under environmental stress conditions, and to assess its viability as a time-dependent forensic marker. *Synechocystis* 6803 cells were cultured under control and high-temperature conditions and were examined over time to determine how their growth, morphology and metabolite overflow products. Cells and growth medium were examined by measuring OD, microscopic studies, Fourier-transform infrared spectroscopy (FTIR), and high performance liquid chromatography (HPLC) analysis. FTIR measurements were used to analyze the cells, and HPLC approach was used to analyze metabolites in the growth medium. HPLC profiles are available to monitor the changes in extracellular metabolite levels resulting from growth and energy utilization. Characterization of the biochemical changes in the cells was performed using (FTIR), and the changes in the cellular components were evaluated with the extracellular metabolite profiles. HPLC and FTIR could give biochemical signatures of the growth and environmental stress of the cells, because this time dependence is observed. Changes in photosynthetic activity, cellular metabolism, and energy balance under elevated temperatures may affect the composition and amount of metabolites excreted by cyanobacteria. These alterations can then be measured as a sign of a change in physiological state and exposure time. Previous investigations of time-dependent degradation of cocaine and norcocaine under environmental conditions support the forensic significance of this work (Chappell et al 2020). Temporal information and information on cyanobacterial changing metabolite profile might also be useful to estimate the forensic time interval similar to chemical degradation. Additional HPLC and FTIR analyses, combined with chemometric techniques such as principal component analysis (PCA) and partial least squares regression (PLS), will be used to detect pattern changes to help characterize growth stages and environmental exposures through future work. This research set out to define an energy management mechanism that can be measured, the cyanobacterial metabolite overflow, and investigate the possibility of it serving as a new biological/chemical forensic time interval estimation tool. This work is supported by the Alabama State University and, in part, by the U.S. Department of Energy, RENEW program (DE-SC0024702).

Jacob Sebesta, Michael Cantrell, Eric Schaedig, Harvey J.M. Hou, Colleen Pastore, Katherine J. Chou, Wei Xiong, Michael T. Guarnieri1, and Jianping Yu* (2024) Polyphosphate kinase deletion increases laboratory productivity in cyanobacteria. Front. Plant Sci. 15:1342496.

Deshayla Chappell, Anthony Asonze, and Harvey J.M. Hou* (2020) Degradation of Norcocaine: A GCMS and FTIR Study, Front. Sci. Technol. Eng. Math., 4, 66-75

Reprogrammed Carbon Partitioning Drives Enhanced Photosynthesis in *Arabidopsis sota* Mutants

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The *suppressor of tyra2* (*sota*) mutants of *Arabidopsis thaliana* carry point mutations in 3-deoxy-D-arabino-heptulosonate 7-phosphate synthase (DHS), the entry enzyme of the shikimate pathway. These mutations relieve feedback inhibition by aromatic amino acids (AAAs) and their derived metabolites, resulting in hyperaccumulation of Phe, Tyr, and Trp and a ~30% increase in net CO_2 assimilation (*A*). Whether this enhanced photosynthesis is driven by increased sink demand for AAA biosynthesis or by reprogramming of central carbon metabolism, including the Calvin-Benson-Bassham cycle, remains unknown.

To address this, we combined ^{13}C-metabolic flux analysis with ^{14}C-based measurements of carbon partitioning in wild-type (Col-0) and two *sota* mutants (*sotaA4* and *sotaB4*). Our ^{13}C-MFA workflow combines short-term $^{13}CO_2$ labeling with fast-quench sampling, LC-MS measurement of mass isotopomer distributions, and computational modeling to generate flux maps. ^{14}C labeling was used to quantify carbon partitioning into starch, sucrose, and shikimate/AAA pools. Contrary to the sink-demand hypothesis, the increased photosynthetic carbon gain in *sota* mutants is not primarily partitioned to sucrose or to shikimate/AAA pools. Instead, most of the newly fixed carbon is directed to starch. The increase in *A* is substantially greater than the increase in shikimate and AAA pools, and long-term ^{13}C labeling indicates that biosynthetic flux to shikimate and AAAs is 2-5 orders of magnitude lower than the increase in *A*. These findings argue against a simple carbon mass-balance explanation and suggest that increased AAA sink demand is not the primary driver of enhanced photosynthesis.

Relative abundance analysis shows that erythrose-4-phosphate and phosphoenolpyruvate, the two substrates of DHS, are significantly elevated in *sotaA4* and *sotaB4* relative to Col-0. Several Calvin-Benson-Bassham cycle intermediates (3-PGA, FBP, G6P, S7P) are also higher in the mutants. These metabolic signatures suggest a reprogramming of central carbon metabolism that may increase the capacity for CO_2 fixation. These studies will provide new insight into the relationship between AAA biosynthesis, carbon partitioning, and photosynthetic regulation, with implications for engineering carbon partitioning toward high-value aromatic compounds.

Anionic Lipid Substitution Shapes LH1–Reaction Center Organization and Function

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Photosynthetic membranes contain a mixture of neutral galactolipids and anionic lipids, including phosphatidylglycerol (PG) and sulfoquinovosyldiacylglycerol (SQDG). In response to nutrient limitation, photosynthetic organisms can adjust their membrane composition by replacing PG with SQDG during phosphate starvation, or SQDG with PG under sulfur limitation. This lipid substitution suggests that maintaining an appropriate pool of anionic lipids may be more important than the identity of the individual lipid species. How these alternative anionic lipids interact with photosynthetic membrane proteins, however, remains poorly understood.

Here, we use coarse-grained molecular dynamics simulations with the Martini force field to examine the effects of PG and SQDG on monomeric and dimeric light-harvesting complex 1–reaction center (LH1-RC) complexes from purple photosynthetic bacteria. Despite their distinct chemical headgroups, PG- and SQDG-rich membranes exhibit similar overall physical properties. In both membrane environments, LH1-RC preferentially associates with anionic lipids over neutral lipids, with this enrichment particularly evident around the protein. Anionic lipids in the inner leaflet form persistent interactions with basic residues near the LH1–RC interface, suggesting a role for lipid–protein contacts in organizing the complex and facilitating excitation-energy transfer from LH1 to the reaction center. These interactions are maintained with both PG and SQDG, despite differences in their headgroup chemistry.

Together, these results show how chemically distinct anionic lipids can provide similar membrane-level interactions with LH1-RC and support the idea that PG and SQDG can functionally substitute for one another while preserving the anionic character required for photosynthetic membrane organization.

Chloride-For-Bicarbonate Substitution Affects Multiple Sites of Activity in Photosystem II

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Bicarbonate is a cofactor of photosystem II (PSII) that regulates various light-driven electron and proton transfers necessary for photosynthesis. In *Limnospira maxima*, a hypercarbonate-requiring cyanobacterium, it is present at multiple sites of PSII, most notably in coordination to a non-heme iron located between plastoquinones Q_A and Q_B. Here, we provide further validation of theorized bicarbonate functions on both the donor and acceptor sides of PSII via *in vivo* depletion of *L. maxima*. Removal was accomplished by exposure to chloride, which readily exchanges with bicarbonate at low-specificity anion sites, and arginine, a known bicarbonate chelator. Lower oxygen production rates and gradual loss of oscillations observed in chlorophyll fast repetition rate fluorometry confirmed deactivation of a significant fraction of water-oxidizing complex centers upon depletion, which was partially reversed after reintroduction of bicarbonate and re-exposure to light. Q_A^- reoxidation kinetics suggest increased charge recombination in depleted cells yet show an increase in efficient transfer to Q_B in a larger fraction of centers when repleted with bicarbonate. Relative chlorophyll and phycobilin fluorescence emission at 77K suggests that phycobilisome attachment rates to reaction centers were not affected by the depletion. This observation differs from formate-based bicarbonate depletion methods that report detachment of the antenna, implying that substituted chloride could retain the speculated function of bicarbonate in phycobilisome association. Absorbance kinetics of downstream electron transport chain components indicate highly reduced plastocyanin despite P700 remaining highly oxidized, signifying electron transfer hindrance at this site in absence of bicarbonate.

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Useful Information

The conference will be held at the Turkey Run Inn, just inside Turkey Run State Park in Marshall, Indiana October 23-25. Meals will be in the main dining area, near the check-in desk. Talks, coffee breaks, and posters will be in the lower level, past the cozy lobby and down the stairs in the meeting area. Forewarned is forearmed, the Wi-Fi can be spotty at Turkey Run, so please be prepared with flash drives to transfer presentations as needed.

Code of Conduct

The Midwest/Southeast Photosynthesis Conference is committed to providing a welcoming, inclusive, and harassment-free environment in all interactions regardless of race, age, ethnicity, national origin, language, gender, gender identity, sexual orientation, disability, physical appearance, political views, military service, health status or religion. We welcome the opportunity to bring photosynthesis research to all people regardless of their identity or background.

As a program that aims to share ideas and freedom of thought and expression, it is essential that the interaction between participants take place in an environment that recognizes the inherent worth of every person by being respectful of all. All participants strive to be empathetic, respectful, welcoming, friendly, and patient. We strive to be collaborative and use language that reflects our values.

The conference does not tolerate harassment in any form. Harassment is any form of behavior intended to exclude, intimidate or cause discomfort. Harassment includes, but is not limited to, the use of abusive or degrading language, intimidation, stalking, harassing photography or recording, inappropriate physical contact, and unwelcome sexual attention.

Examples of unacceptable behavior

All participants are committed to making participation in this community a harassment-free experience.

We will not accept harassment or other exclusionary behaviors or actions that are illegal, such as:

- The use of sexualized language or imagery
- Excessive profanity (please avoid curse words; people differ greatly in their sensitivity to swearing)
- Posting sexually explicit or violent material
- Violent or intimidating threats or language directed against another person or group
- Inappropriate physical contact and/or unwelcome sexual attention or sexual comments
- Sexist, racist, or otherwise discriminatory jokes and language

- Trolling or insulting and derogatory comments
- Written or verbal comments which have the effect of excluding people on the basis of membership in a specific group, including level of experience, gender, gender identity and expression, sexual orientation, disability, neurotype, personal appearance, body size, race, ethnicity, age, religion, or nationality
- Public or private harassment
- Continuing to initiate interaction (such as photography, recording, messaging, or conversation) with someone after being asked to stop
- Sustained disruption of talks, events, or communications, such as heckling of a speaker
- Publishing (or threatening to post) other people's personally identifying information ("doxxing"), such as physical or electronic addresses, without explicit permission
- Other unethical or unprofessional conduct
- Advocating for, or encouraging, any of the above behaviors

All participants are governed by local laws, including Turkey Run State Park rules and their organization's code of conduct and policies.

How to Submit a Report

If you feel your safety is in jeopardy or the situation is an emergency, contact local law enforcement before making a report to the conference organizers. (In the U.S., dial 911.)

Anyone who experiences, observes or has knowledge of threatening behavior is expected to immediately report the incident to a member of the event organizing committee, state park staff, or a trusted friend or colleague. The Midwest/Southeast photosynthesis conference reserves the right to take appropriate action.

Take care of each other. Alert the organizers if you notice a dangerous situation, someone in distress, or violations of this code of conduct, even if they seem inconsequential. For possibly unintentional breaches of the code of conduct, you may want to respond to the person and point out this code of conduct (either in public or in private, whatever is most appropriate). If you would prefer not to do that, please report the issue to the conference chair or co-chair.

What Happens Next

All complaints will be reviewed and investigated and will result in a response that is deemed necessary and appropriate to the circumstances. All reports will be kept confidential, with the exception of cases where the organizers or state park staff determines the report should be shared with law enforcement. In those cases, the report will be shared with the proper legal authorities.

In some cases the conference organizers may determine that a public statement will need to be made. If that's the case, the identities of all involved parties and reporters will remain confidential unless those individuals instruct us otherwise.